

Degree Project in Innovation Management and Product Development
Second Cycle, 30 credits

Early Access Development in the Gaming Industry

An exploratory study

FAREED EFFAN

ANTONIO GARGIULO

Early Access Development in the Gaming Industry

An exploratory study

Fareed Effan
Antonio Gargiulo

Master of Science Thesis TRITA-ITM-EX 2024:167
KTH Industrial Engineering and Management
Innovation Management and Product Development
SE-100 44 STOCKHOLM

Examensarbete TRITA-ITM-EX 2024:167

Early Access utveckling inom spelbranschen: En utforskande studie

KTH Industriell teknik och management

Fareed Effan
Antonio Gargiulo

Godkänt	Examinator	Handledare
28 maj 2024	Sofia Ritzén	Mats Magnusson
	Uppdragsgivare KTH Royal Institute of Technology	Kontaktperson Chris Toepker

Sammanfattning

"Early Access" hänvisar till en mjukvarulanseringsstrategi där konsumenter kan skaffa sig en ofullständig version av mjukvaran och bidra till utvecklingsprocessen genom att ge feedback. Det har blivit en framträdande preferens inom branschen, ett växande antal spelutvecklare väljer att samutveckla sina kommande spel tillsammans med sin kundbas. Även om strategin är allmänt använd, är uppfattningen om vad termen "Early Access" praktiskt översätts till fragmenterad: de operationer och metoder som omfattas under etiketten skiljer sig från en studio till en annan, vilket skapar otydlighet kring termen och saknar en klar, gemensam förståelse för de principer, variationer och ramverk som kan användas för att implementera denna strategi. Denna forskning syftar till att skapa klarhet kring modellen, förstå dess viktigaste element, vanliga metoder och processer. Vi syftar till att ge insikter om framgångsfaktorer och utforska deras dynamik från både utvecklarnas och gemenskapens,en grupp av engagerade konsumenter, perspektiv på processen. Vi genomförde en forskningsstudie, baserad på resultaten från en kvalitativ analys av insikter som samlats in från 13 semistrukturerade intervjuer med utvecklare, community managers och experter inom branschen. Analysen följde GIOIA-metodologin. Detta gjorde det möjligt för oss att avveckla den intrikata uppsättningen av insikter och strukturera informationen i sex huvudaggregerade dimensioner av de centrala elementen i "Early Access". Dessa dimensioner jämfördes sedan med resultaten från en litteraturöversikt av artiklar som beskriver dynamiken och praxis som delar samma principer för Early Access” metoder. Resultatet är en omfattande analys av ”Early Access” metodologi, liksom en uppsättning av principer och nyckelfaktorer för en framgångsrik applicering av modellen. Slutsatsen sammanfattar de grundläggande principerna “Early Access”-modellen, klargör dess tvetydiga definition och identifierar variablerna som skapar variationen i implementeringen av “Early Access” samt deras samspel. Gemenskapsbyggande element och dess nyckelfaktorer för framgång analyseras och en utvidgad strategi för feedbackloopen presenteras, där vikten av en vision under utvecklingsprocessen och utvecklingen av roller och strukturer till följd av implementeringen av modellen lyfts fram.

Master of Science Thesis TRITA-ITM-EX 2024:167



KTH Industrial Engineering and Management

Early Access Development in the Gaming Industry
An Exploratory Study

Fareed Effan
Antonio Gargiulo

Approved	Examiner	Supervisor
May 28, 2024	Sofia Ritzén	Mats Magnusson
	Commissioner KTH Royal Institute of Technology	Contact person Chris Toepker

Abstract

"Early Access" refers to a software release strategy wherein consumers can acquire an incomplete version of the software and contribute to the development process by providing feedback. It has developed into a prevalent preference in the industry, embraced by a growing number of studios that decide to co-develop their upcoming games together with their customer base. Although the strategy is widely used, the perception of what the “Early Access” term practically translates to is fragmented: the operations and practices encompassed under the label differ from one studio to another, generating ambiguity around the term and lacking a clear, common understanding of the principles, variations, and frameworks that can be employed to implement this strategy. This research aims to build clarity around the model, understanding its key elements, common methods, and processes. We aim to provide insights into success factors and explore their dynamics from both the developers' and community perspectives of the process.

We performed an exploratory study, based on the findings from a qualitative analysis of insights gathered from 13 semi-structured interviews with developers, community managers, and practitioners within the industry. The analysis followed the GIOIA methodology, which enabled us to unfold the intricate array of insights and structure the information into six main aggregate dimensions of the core elements of Early Access, which were then compared to the results of a literature review of articles related to fields that share similar dynamics with Early Access practices.

The result is an extensive analysis of the Early Access methodology, as well as a set of principles and key factors for a successful approach to the model. The conclusion incorporates the pillars of the Early Access model, decoding its ambiguous definition, and presenting the variables that generate the variation of Early Access implementations, and their interdependencies. Furthermore, we analysed the community-building element and its key success factors, as well as presented an expanded approach to the feedback loop, highlighting the importance of a vision during the developed process and the evolution of roles and structures due to the method implementation.

FOREWORD

Firstly, we express our sincere appreciation to our supervisor Mats Magnusson, whose guidance has been instrumental throughout this academic journey. Additionally, heartfelt gratitude is extended to our industry contact, Chris Toepker, who has been crucial in significantly enriching the practical dimension of this research with his expertise and support. Hence, we would like to thank the studios that participated in the interviews, providing invaluable data and insights that have enriched both the theoretical framework and practical dimensions of the research.

A particular acknowledgement is reserved for DJTruthSlayer from Tinybuild for his generous contribution of the "User Sentiment Report", and Chris Bourassa, Co-Founder of RedHookStudios for finding the time to share key insights from his experiences with Early Access development.

Furthermore, we wish to convey our heartfelt appreciation to KTH Royal Institute of Technology for providing a conducive academic environment and the resources necessary for the completion of this thesis. Ultimately, a special thanks goes to all the participants and contributors to the research, as well as the practitioners and developers in the industry for creating incredible games.

NOMENCLATURE

EA: Early Access release model (Acronym)
Players: refer to individuals who engage with and participate in playing video games, interacting through various gaming platforms, such as consoles, personal computers, mobile devices, or dedicated gaming systems.
Modding: Video game modding (short for "modification") is the process of alteration by players or fans of one or more aspects of a video game, such as how it looks or behaves. (Poor, 2014)
Agile Development: The Agile methodology is a set of frameworks, values, and principles designed to guide and improve how development teams work together, based on the Manifesto for Agile Software Development.
Player Community: A group/network of individuals sharing a mutual interest in a specific game, which they play regularly, or being invested in a specific studio's production, often offering technological and informational support to tackle complex game mechanics. (Behrenshausen, 2013)
Studios: Companies and enterprises whose main activity is game development.
Late-game content: In the context of video games, "late game" refers to the advanced or final stages of gameplay. It is a phase where players have progressed significantly in the game and frequently engage in activities that require the culmination of skills and resources gathered throughout the earlier parts of the game.
APIs: An application programming interface (API) is a way for two or more computer programs to communicate with each other. It is a type of software interface, which offers a service to other pieces of software. (Reddy, 2011)
AAA: (pronounced "triple-A") refers to a classification of high-budget and high-quality games produced by major game development studios or publishers. These games typically have large production teams, significant financial investment, and extensive marketing campaigns.
Release State: the specific stage of development that a game has reached when it is made available to the public for purchase, download, or play. It indicates the version or state of the game that players experience.

TABLE OF CONTENTS

1. INTRODUCTION _____ *1*

1.1. Background _____ **1**

1.2. Purpose _____ **2**

1.3. Study Scope and Limitations _____ **2**

2. LITERATURE REVIEW _____ *3*

2.1. Introduction to the Gaming Industry _____ **3**

2.1.1. The Early Access Development Model _____ 3

2.2. Swedish Contributions to Gaming _____ **4**

2.3. Software Development Processes _____ **4**

2.4. Co-Creation and Collaborative Development _____ **6**

2.4.1. Examples from within the Gaming Industry _____ 6

2.4.2. Managing Communities in Software Development _____ 7

2.4.3. Impact of Social Software on User Communities _____ 9

2.4.4. Open-Source Development and Customer Involvement _____ 10

2.5. Leveraging the "Crowd" Element _____ **11**

2.5.1. Crowdsourcing _____ 11

2.5.2. Crowdfunding _____ 11

2.6. Literature Summary _____ **12**

2.6.1. Proposed Factors Based on Literature _____ 14

3. METHODOLOGY _____ *15*

3.1. Data Collection Process _____ **15**

3.1.1. Demographic Summary of Interview Candidates _____ 16

3.2. Analysis Method _____ **16**

4. RESULTS AND ANALYSIS _____ *18*

4.1. Introduction to Results _____ **18**

4.2. Early Access - An Expansive Label _____ **18**

4.2.1. Historical Challenges with Early Access _____ 18

4.2.2. Impact of Release State _____ 20

4.2.2.1. Zone A – Risk of Unplayability _____ 21

4.2.2.2. Zone B – Misuse of EA Label by AAA Studios _____ 21

4.2.2.3. Zone C – Extreme focus on Quality _____ 21

4.2.2.4. Zone D – Prioritising Content Quantity _____ 21

4.2.2.5. Zone E – Balanced Content Quality and Quantity _____ 21

4.2.3. Multiple Phases of Availability _____ 22

4.3. Clarity of Vision and Its Importance _____ **22**

4.4. Multiple Variations of Early Access _____ **25**

4.5. Community Building is a Core Element _____ **27**

4.5.1. Maintaining Transparent Communications _____ 28

4.5.2. Humanisation of Development Teams _____ 28

4.5.3. Importance of Consistent Communication _____ 29

4.6. Expanded Approach to Feedback _____ **30**

4.6.1. Evolved Feedback Loop 31
4.6.2. Methods for Information Dispersion 32
4.6.3. Feedback Collection Methods 33
4.7. Evolution of Processes and Roles Due to Early Access 34
4.8. Proposed Updated Model 35
5. DISCUSSIONS & CONCLUSIONS 36
6. RECOMMENDATIONS AND FUTURE WORK 38
6.1.1. Recommendations 38
6.1.2. Future Work 38
7. WORKS CITED 39
8. APPENDIX 1
8.1. User Sentiments Report 1
8.2. Interview Questions 4
8.3. Candidate Summary 6

1. INTRODUCTION

In recent years, the gaming industry has become one of the dominant sectors within digital media, with an estimated market size of over 200 billion USD seen in 2022 (Grand View Research, 2023). While multiple factors have led to this scale, one of the key aspects is the continual focus on innovation and new developments within the industry.
This persistent emphasis on chasing new areas of improvement often results in every aspect and system within the industry undergoing some form of evolution, covering everything from technical advancements that enable new methods of development and delivery, to pushing creative boundaries in building innovative new experiences, all the way through to new business models (Landoni, Dell'era, Frattini, & Petruzzelli, 2020) that open new avenues for appropriating value.

1.1. Background

One such innovation has been the Early Access development model (Sandqvist, 2015), which has gained significant traction within the industry over time and has grown to become a prominent choice that is frequently selected by an increasing number of studios as they approach their next game. An empirical analysis of one game marketplace and distribution platform Steam in 2017 showed that over 15% of the games on Steam actively use, or at some point leveraged, the Early Access model and its popularity is still growing. (Lin, Bezemer, & Hassan, 2017)
The core characteristic of this model is that it creates a unique scenario wherein the player base is significantly more engaged throughout the overall process where they are enabled to directly influence, and sometimes even contribute significantly to, the development of the software and direction of the game. This creates a unique approximation of an outside-in open innovation framework (Bagherzadeh, Markovic, Cheng, & Vanhaverbeke, 2019) that creates a bridge between the development teams and the end users. Hence, this iteration allows for a collaborative effort that can have a significant impact on the final value delivered to the players. This can further be bolstered by elements of crowdfunding and pre-sales that serve to alleviate some of the financial pressures seen within a highly competitive industry known for innovation across both technical and creative competencies (Plum & Hassink, 2014).
While such a model can have a very positive impact on the overall industry by alleviating the burden of growing development costs and lowering the barrier for more contenders to enter the domain, there is still an element of risk involved as with any product development cycle. This is particularly true when discussing Early Access games which involve a core element of releasing an unfinished product to the public, which can sometimes have damaging consequences for any studio or player base (Mullen, 2021).

1.2. Purpose

Our research aims to act as an exploratory study into the Early Access development model, aiming to generate a better understanding of the model and its key elements. The focus areas we intend to explore are which methods and processes are commonly used within an Early Access development cycle, along with the unique factors of this model that enable the creation of more innovative experiences that are closely aligned with the needs of the player base.
Expanding on this further, our goal is to share some insight into some of the common success factors for such a model, as well as its relationship with the community element of the player base.

1.3. Study Scope and Limitations

Taking into perspective the tremendous scale of the game development industry, it is pertinent to define the scope of our study and highlight the areas of the industry that we are focusing on. Since most Early Access initiatives are focused on releases targeting console and computer games, our study excludes the mobile game and app development segments of the industry.
Alongside this, it is important to address the limited commercial data and publicly accessible information around game development business models, particularly Early Access, as it limits the scope of our analysis. One such example is that only one of the major game distribution platforms (Steam) has publicly addressable APIs to access information about player insights and game performance. This also means that closed services with limited public insights such as subscription access models (Xbox Game Pass) are also excluded from the study due to the limited public information about their scale and performance.
Finally, while our study is based on 13 interviews with community managers, studios, and publishers, it's important to acknowledge that within the massive scale of the industry, this is still a relatively limited data set and further interviews may yield additional insights. One must also consider that it is still difficult to account for factors that could drive failed product cycles as that can often lead to defunct companies that are complicated to trace.

2. LITERATURE REVIEW

The initial segment of our literature review provides an initial introduction to the Early Access model while also offering some additional insights into the broader gaming industry. From an internal perspective, studios implementing Early Access often introduce organizational dynamics that align with principles of software development and agile methodologies. Keeping this community-driven nature in mind, we expanded our research to cover some insights on co-development and co-creation and the common methods associated with them. Moreover, recognizing the significance of community management and the financial dimensions bound to it, we draw valuable insights from studies on managing online communities in software development, as well as the frequently utilised elements of crowdsourcing and crowdfunding. Lastly, our review extends its scope to encompass the aspects of open innovation and practices suited for open-source development.
These insights and findings from literature are then combined into a rough model that can be used as a comparative starting point for software development scenarios where these factors are either present or proactively being introduced.

2.1. Introduction to the Gaming Industry

Expanding on the insights shared in the introduction about the scale of the video game industry, forecasts show that the trends for growth will continue well into the future with revenue from this segment expected to increase to above 300 billion USD by 2027 (Statista Market Insights, 2023). One study in China also showcased the quantifiable knock-on effect the game industry has on the overall GDP of the country, leading to, highlighting the rapidly growing esports segment as a key driver for this growth (Bai, Chen, Shang, & Wang, 2022).
Aside from being an economic behemoth, the prevalence of video games in modern culture is also an important consideration due to their significant contributions to how society has developed over time (I.V. & M.V., 2021). The most recent example of this phenomenon was the growth in popularity and immense growth seen throughout the industry during the COVID-19 pandemic, where online games played a key role in bridging the social gap created by distancing restrictions (Bai, Chen, Shang, & Wang, 2022).

2.1.1. The Early Access Development Model

The gaming industry has often been at the forefront of innovation in software, pioneering new methods and initiatives that are frequently adopted by other industries. One relatively recent innovation in this list is the Early Access development method. This approach to game development entails an interactive process between developers and their user base, based on a feedback loop aimed at fostering collaboration and often leading to co-developing the product. Despite a scarcity of academic research on the subject, we observed numerous parallels with different disciplines and fields, warranting further investigation for the purpose of our study.

2.2. Swedish Contributions to Gaming

Before proceeding to more detailed topics around the development processes surrounding software development, on key acknowledgement is the history of Swedish contributions to the global gaming landscape. Among the different "clusters" for game development, Sweden has perhaps the most prolific national presence within Europe. In the past year (2023) alone, games developed in Sweden saw an impressive seven billion downloads, with major titles often seen leading the global sales lists. Even across some difficult years for the gaming industry, the revenue for the Swedish gaming industry has seen a 13% increase bringing the total revenue to 3.1 billion Euros in 2023.
Expanding this to include foreign subsidiaries further increases this figure up to 8 billion Euros, which is a 40% increase compared to previous years. As stated in a recent report by Dataspelsbranschen, a national association that represents and monitors the Swedish gaming industry: "Every tenth person has played a game created in Sweden". Most of the Swedish game companies are profitable, and the industry contributes several billions to the public treasury. Together, the ten most profitable companies paid about EUR 250 million in corporate taxes on their profits in 2022.

2.3. Software Development Processes

As with any industry pertaining to software, the processes and methods used throughout the development are a major factor that drives the way work is organised and carried out. Looking at the broader software industry, the framework that has proliferated to the point of becoming commonplace is the Agile development methodology (Beedle, Bennekum, & Cockburn, 2001). The frequency with which this framework is chosen is quite understandable however, providing four core principles accompanied by a set of guidelines and that are fluid enough to be enacted across a wide range of scenarios. (Alsaqqa, Sawalha, & Abdel-Nabi, 2020)

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

Figure 2.3-1: Core Principles of The Agile Framework (Beedle, Bennekum, & Cockburn, 2001)

These core principles, and the relatively open approach taken when implementing them, has led to various innovations and process adaptations over the years that enable development teams to more closely follow the core ideology of Agile. Chief among such innovations are the ones that came about due to the emphasis on reactivity, leading to an increased focused on testing (Najihi & Elhadi, 2022) and the use of an iterative approach (Larman, 2004), both for design and development practices.
The Agile framework's widespread adoption and success has even crossed digital boundaries, frequently being implemented to manage projects across a wide range of industries (Denning, 2013). The focus on close collaboration with the customer allows for a working approach that is significantly more conducive to continuous innovation to meet evolving user needs. However, the same ambiguity that has made the Agile framework such a widely adopted option is also the cause perhaps it's biggest pitfall, with an increasing number of conflicting definitions and implementations causing confusion across both the academic and industrial worlds (Abrahamsson, Salo, & Ronkainen, 2002).

In comparison, exploring the core processes and methods that keep the gargantuan gaming industry afloat, showed a wide range of development methods being employed across the field (McKenzie, Trujillo, & Hoermann, 2019) ranging from more conventional software development methods through to modified variants of agile frameworks. Looking further into this mention of agile shows an interesting trend, as many principles of the framework can often be misapplied or misunderstood by game development companies (McKenzie, Morales-Trujillo, Lukosch, & Hoermann, 2021). Overall, given the exceedingly complex need to combine competencies across both software engineering and creative production, the lack of any one widely accepted framework or methodology for video game development becomes more explicable.

This same ambiguity and absence of well-defined best practices and frameworks also extend through to the Early Access model, with different variations and implementations seeing fluctuating degrees of success. Compounding this challenge is a notable absence of previous literature within the area, with prior studies primarily focusing strictly on elements such as the growth of the model (Lin, Bezemer, & Hassan, 2017), evolving user perceptions (Arafat, Qusef, & Al-Taher, 2019), or economic outcomes (Consalvo & Paul, 2018). Moreover, some of these studies have also painted a relatively bleak picture of the model, suggesting that the model has evolved to being nothing more than buggy, incomplete, or abandoned games coexisting with a slew of broken promises (Mullen, 2021).

2.4. Co-Creation and Collaborative Development

With the introduction of technology and the various means of communication, the process of innovation is becoming democratized, as people are increasingly capable of driving the process themselves. This necessitates companies to continuously challenge their existing processes and reinvent their businesses as they work to adapt to modern challenges and the accompanying dynamic and fragmented market landscapes. (Rubio, Villaseñor, & Yagüe, 2021)
In today's rapidly changing business climate, co-creation is quickly emerging as a core strategy for such success and innovation. Co-creation is considered as a trend situated to social or mutual achievements and can also be seen as an opportunity tailored for firms looking to better address the needs of their consumers. The customer co-creation concept refers to the strategy of including customers in the process of product improvements and the development of new offerings (Prockl, Bhakoo, & Wong, 2017), wherein value is created through a collaboration between the company and the end customer (Priante & Ehrenhard, 2017)
In their path towards obtaining a robust competitive advantage closely tied to a compelling value proposition, firms are driven to innovate by fostering elevated levels of participation through an effective co-creation approach (Cornelio & Cruz, 2014). In their research, (Iglesias, Ind, & Alfaro, 2013) explores the impact on brand value from companies engaging in such co-creation efforts with consumers, emphasizing how this practice not only fosters fruitful and effective innovations but also opens new opportunities for the business.
Furthermore, it is essential to develop and utilize platforms to foster the co-creation process, to combat information overload, and keep consumers notified about what is happening to their idea. (Wadeisa & Sada, 2015)'s findings highlighted that firms are challenged and find difficulty in integrating knowledge between them and their consumers.

2.4.1. Examples from within the Gaming Industry

While there is a wealth of information around co-creation and collaborative development available for other software related industries, there is notable lack of academic research around the collaborative aspects within game development and the Early Access model. To obtain an initial reference about collaborative development, one of the core aspects of Early Access, we expanded our search to cover the other verticals within game development to gather insights from past studies in collaborative game design and development.
One study focused on mobile games investigated the benefits, in both commercial elements as well as innovation, of developing specific toolkits to enable consumers to contribute without needing specialised technical knowledge in the field (Piller , Ihl, & Füller, 2004). A further study into open-sourced development within the creative industry provided a showcase of private-collective innovation and how it enabled firms within the creative industry to include consumers more openly in their efforts while still being able to appropriate value from their products (Erickson, 2018).

2.4.2. Managing Communities in Software Development

Fostering and managing online communities has been an important focus area for scholars researching the topic of co-creation alongside customers. (Constantinides, Brünink, & Romero, 2015); (Priharsari, Abedin, & Mastio, 2020). Given the fundamental need for customer involvement in any co-development process, the ability to effectively navigate and manage online communities is a critical success factor for generating value and fostering innovation through this approach.

To accomplish this monumental task, companies need to gather insight into different factors that drive customer motivation and involvement to master the art of attracting and involving the ideal individuals to their projects (Constantinides, Brünink, & Romero, 2015). To be able to recruit the ideal user segments as collaborators, businesses must put an increased emphasis on understanding the motives and incentives that could drive them to contribute to firm-level open innovation activities (Nohutlu & Englis, 2023).

Example motivation	**Example customer behavior**	**Example company practice**
Feeling a part of the project Feeling of self-efficacy	Taking initiatives in organizing the co-creation process	Give more responsibilities to the community member and autonomy in those tasks, such as creating posts for community to inspire knowledge and idea sharing
Satisfaction that arises from improving current products/contribution with new product or service ideas	Proposing solutions to improve current product/service-related issues	Invite the community members for a brainstorm with company representatives (e.g., product owners, program managers) to provide improvement inspiration
Gaining problem-solving ability	Getting involved in the process of creating solutions to problems, collaborating with others and learning	Introduce challenges for team or individual submissions, enable voting and recognition with a top innovator prize
Interacting with like-minded others Developing a sense of belonging/community among community members	Encouraging knowledge and feedback sharing and relationship building within the community	Start a communication campaign for “Community connector/liaison” and position the community task/involvement as an opportunity to build connections and facilitate communication among others
Learning opportunities	Learning through individual efforts and from others’ contributions	Facilitate feedback sharing and building on community member’s contribution
Altruism	Showing care, attention, and help toward the co-creation project and the community	Create encouragement in voting and supporting others’ contributions that will help to motivate new submissions and select top ideas

Figure 2.4-1: Examples of Customer Behaviours and Company Practices (Nohutlu & Englis, 2023)

In exploring some of these motives, (Nohutlu & Englis, 2023) identifies and creates an overall taxonomy around the motivations and benefits consumers gain from participating in co-creation activities. This research underscores the significance of grasping both extrinsic motivations (driven by goal achievement) and intrinsic motivations (arising from a genuine interest in the activity) among consumers throughout the collaborative innovation process.

Throughout this process of collaborating to innovate towards new products, companies often share transparent information with their customers and involve them by listening to them, and therefore, customers show trust and loyalty toward their choice brands (Urban, 2005). This initial effort can create a cyclical effect, wherein companies and customers become advocates or "champions" of each other, resulting in a combined increase in actual and perceived value. As an example, customers driven to collaboratively create value can effortlessly express their needs, provide feedback, and share information through Internet-mediated channels.
One key research effort was the study carried out by (Prahalad & Ramaswamy, 2004), wherein they investigated several online communities to develop their DART model which introduces four core pillars that are needed when working to create value among any community. For this model, the four pillars cover simple guidelines that emphasise the need for dialogue, access, risk assessment and transparency. For instance, "dialogue" represents the engagement and inclination to act from both sides, which in our case translate to the relation between developers and community.

Access and transparency. Coupling access with transparency enhances the consumer's ability to make informed choices.	**Dialogue and risk assessment.** Combining dialogue with risk assessment enhances the ability to debate and co-develop policy choices.
Access and dialogue. Coupling access with dialogue enhances the ability to develop and maintain thematic communities.	**Transparency and risk assessment**. Combining transparency with risk assessment enhances the ability to co-develop trust.

Figure 2.4-2: Combinations of the DART Model (Prahalad & Ramaswamy, 2004)

2.4.3. Impact of Social Software on User Communities

With the prevalence of social media platforms in the modern world, their impact on online communities is an important consideration to investigate. These social media platforms facilitate and often strengthen the dynamic interactions within online communities, making it possible for customers to seamlessly share stories about interactions and experiences with brands with others. (Hajli, Shanmugam, Papagiannidis, Zahay, & Richard, 2017)



Figure 2.4-3: Recursive Process of Branding Co-Creation (Hajli, Shanmugam, Papagiannidis, Zahay, & Richard, 2017)

These social media platforms and their adjacent online communities are often important spaces for interacting closely with customers, contributing jointly with the company to the co-creation process. In the context of these virtual communities, even participants focused on personal benefit contribute to the co-creation of value as they assist in designing an experience tailored to their wants and needs. Through this communal approach to value creation, users often end up creating value in the form of evaluations, recommendations, and ideas that are beneficial to others as well.

Also, social media platforms and other online communities are important spaces for interacting with customers, as they contribute jointly to the co-creation process (Zwass, 2010). In the context of virtual communities, through self-value co-creation, participants co-create value to their own benefit by becoming involved in co-designing their experience. By communal value co-creation, participants co-create value through evaluations, recommendations, and ideas that benefit others (Rubio, Villaseñor, & Yagüe, 2021).

The same importance of community interaction and management also extends towards the gaming industry where online player communities are a key pillar for any game's success, especially so when looking at a collaborative model such as Early Access. When exploring the history of video games, particularly those with an online or social component, we can see that involved player communities have often held a massively influential role within the entire industry. A study conducted in 2011 that explored this phenomenon specifically found that the player base and its strong interaction with the development firms is a fundamental element of the gaming industry, finding that the impact of such a relationship has increased significantly with a growing global emphasis on social software. (Burger-Helmchen & Cohendet, 2011)

Other interesting studies around this topic highlighted the benefits of social communities (Wiederhold, 2021) in gaming, from the perspectives of inclusivity (Wirman, 2014), as well as their positive impact on learning behaviours (Gandolfi, Ferdig, & Soyturk, 2023). With the key role the player base holds within this space, it's also relevant to consider the factors that contribute to the formation of a strong community. While there is slightly limited literature available within this area, some past articles are quite informative about the roles associated with forming and maintaining strong communities (Salovaara, Johnson, Toiskallio, & Tiitta, 2005).

2.4.4. Open-Source Development and Customer Involvement

One key example of collaborative development that has quickly become commonplace within the software industry is the concept of open-source development, which has a community-driven approach to software creation as its core. In terms of the conventional software industry, this is the model that draws the most parallels with Early Access, as they both embody a fundamental departure from the traditional closed development methods. The collaborative nature of open-source projects mirrors the growing emphasis on user base participation during the Early Access stages of game development. The symbiotic relationship seen in open-source projects between developers and end-users not only accelerates the iterative process but also fosters a sense of community ownership and shared enthusiasm for the evolving experience.

Over the past few years, the open-source business model has seen increased adoption, with studies highlighting the advantages and successful outcomes associated with creating commercial software using an open-source approach (Krishnamurthy, 2005). Drawing further parallels with Early Access, an open-source model gives the users of a software system the ability to actively participate in the development of the product and contribute their own time and effort on aspects of the product that they find most important.

With recent advances in technology, software projects now have access to new levels of communication between consumers and developers (Abelein, Sharp, & Paech, 2013). Open-source software allows consumers to take an active part in the development process by contributing code, identifying and fixing programming bugs, and determining the end uses of the product (von Krogh, Haefliger, Spaeth, & W. Wallin, 2012). However, involving a community in the development process may require significant changes to a company's structure and development processes, as both external and internal contributions need to be accommodated at the same time (Bettenburg & E. Hassan, 2013)

The need for this continuous integration between external and internal knowledge sources within the open innovation boundaries becomes an important challenge to tackle, as it requires intense knowledge management practices within any product team (Vaisnore & Petraite, 2011). Firms need to be cognisant of these issues and must work to address them when approaching open innovation. A promising approach to this knowledge management challenge was the dynamic community framework proposed by Ye in 2005 (Ye, Yamamoto, & Kishida, 2004), wherein there is a combination of the traditional knowledge management approach in which knowledge is formalized and accumulated in knowledge repositories, and the community-based knowledge collaboration approach in which knowledge is transferred through informal community participation and human contacts.

Similarly, projects focused on open innovation models such as open-source software often require major modifications in organizational structures, goal establishment frameworks, and innovation strategies to facilitate the empowerment of knowledge flows and exchanges among both internal and external contributors (Chesbrough & Crowther, 2006). Moreover, innovation studies across the past decade have increasingly focused on the changes of innovation model, with the emphasis on open innovation processes and their impacts on the firm's innovation performance (Vaisnore & Petraite, 2011)

2.5. Leveraging the “Crowd” Element

Having covered both the processes for collaborative development as well as the aspect of online communities and how they can come together to drive innovation, one additional area to investigate is how to leverage to collaborate across areas outside of the core development practices. While there is a myriad of ways that a company can look to work closely with these pre-emptive crowds, there are two main approaches seen frequently within both the software industry as well as other verticals entirely. These methods can be used either to garner the public’s assistance across a wide range of project tasks (crowdsourcing), or to assist in financing the proposed project where funding options are otherwise limited (crowdfunding).

2.5.1. Crowdsourcing

The term “Crowdsourcing”, initially coined by (Howe, 2006), is the act of an organisation outsourcing their work to an undefined, networked labour using an open call for participation. Through this open invitation, the company can engage a global online workforce to tackle a diverse range of tasks, including everything from ideation to core software development and testing. Advocates of the model frequently assert that this emerging model decreases time-to-market by enhancing parallelism. (Lakhani, Garvin, & Lonstein, 2012). Considering the focus of our study is around game development and the Early Access model, it’s important to note that the domain has expanded to the extent that the classification of ‘Crowdsourced Software Engineering’ has also been defined to denote the application of crowdsourcing techniques to support core software development. In their research, (Mao, Capra, & Harman, 2017) corroborate this by investigating the landscape to find that the most prevalent crowdsourcing application is that of software engineering, among which software design and software testing are also included.

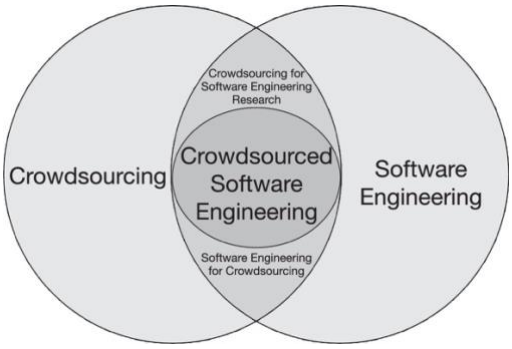


Figure 2.5-1: Intersection between Crowdsourcing and Software Development (Mao, Capra, & Harman, 2017)

Given that one of the commonly touted benefits of Early Access development is the ability to scale testing to a massive scale with user involvement, and interesting parallel can be seen with similar use cases seen for crowdsourcing for testing. This approach has been applied to various types of testing activities, and its advantages based on exploiting a large potential user pool are widely acknowledged. Another interesting result from the same study is the demonstration that crowdsourced usability testing has the capability for detecting usability problems as good as the testing done by ‘experts’ (Schneider & Cheung, 2013).

2.5.2. Crowdfunding

On the opposite end of the spectrum to crowdsourcing, which deals with development activities, crowdfunding pertains solely to the financial aspects of any project. Given that the predominant usage of the Early Access model is by studios on the smaller scale, the financial

aspects are a key area that must be investigated further to better understand the model and the methods surrounding it. Obtaining insights into the funding trends in the gaming industry can often be challenging however, so in this case the initial guidance can also be drawn from other industries to establish a known reference.
Looking to the broader software development landscape, the aspect of "crowdfunding" often refers to a scenario wherein user interest and smaller initial commitment can be leveraged to in effect, bankroll the development of a product or service. An initial exploratory study from 2014 around this phenomenon provides a good explanation of the process, defining the term as a process by which founders can leverage the internet to draw on relatively small contributions from a large base of individuals to fund their design and development efforts. Taking on the empirical data from within the same study also showcases that there is a large bias towards inventive projects being successfully funded over others and that the creative vision is a core driver for users committing to any proposed project. (Mollick, 2014)
Another study from the same year focused specifically on the software development industry, diving deeper into the process and different methods for crowdfunding available within the software industry. The study proposes four main methods for crowdfunding, mainly based on the value traded between the developers and those funding the project. The wide range of transaction types that can fall within this bracket shows that there is a variety of funding options that feed into the label of "crowdfunding", which could be one of the reasons the model has seen such widespread adoption in recent years. (Branzov & Maneva, 2014)

Equity-based	Lending	Donations-based	Rewards-based
Backers receive equities in the company	Backers receive interest over the sum they pledge	Backers donate without expecting any material or virtual rewards in return (however, sometimes fundraisers can decide to express their gratitude)	Backers receive predefined material or virtual goods based on the amount they donate.

Figure 2.5-2: Variants of Crowdfunding (Branzov & Maneva, 2014)

To further expand on the crowdfunding approach and its parallels with Early Access based funding, one must look into other software development-based verticals where similar success has been achieved. A recent study carried out by researchers at DePaul University explores this very phenomenon in more detail, particularly focusing on Open-Source Software (OSS) and the impact it has on the success of a crowdfunding campaign. The study showed that a project that was open source was much more likely to be supported and successfully funded due to it being perceived as a collective effort that anyone can theoretically benefit from. This draws interesting parallels with the collaborative element of Early Access game development, which can often give users the perception that they are more involved in the development process. (Boylan, Nesson, & Philipps, 2018)

2.6. Literature Summary

The exploration of these various fields, and their similarities with relevant dynamics similar to the Early Access model, can offer invaluable insights that contribute to a comprehensive understanding of Early Access practices and methodologies. By examining the parallel dynamics between the broader software industry and the Early Access model seen in the gaming industry, we can extract critical lessons applicable to navigating the complexities of Early Access adoption and management.

For instance, insights from crowdfunding and crowdsourcing shed light on effective engagement strategies and incentivization methods for building and sustaining user communities within the Early Access framework. Similarly, principles from community management provide guidance on fostering active participation and feedback exchange, essential components of successful Early Access initiatives. Moreover, understanding the dynamics of open innovation contributes to identifying opportunities for collaborative development and leveraging external expertise to enhance product quality and innovation within Early Access projects.

Conversely, comprehending the unique characteristics of Early Access dynamics also holds significance for these fields. Insights into the iterative development, rapid feedback cycles, and user-centric approaches inherent in Early Access methodologies can inform strategies for agile development practices, optimizing project management processes for enhanced efficiency and responsiveness. Building on this further, the collaborative nature of co-development in Early Access settings underscores the importance of open innovation principles, emphasizing the value of collaborative ecosystems and knowledge sharing across disciplines.

Thus, this reciprocal relationship between Early Access methodologies and the closely related software development industry reinforces the interlinked nature of modern development paradigms and highlights the multifaceted implications for theory and practice in both academia and industry.

On the other hand, it is necessary to underline the differences and asymmetries between the studied fields and Early Access that warrant careful consideration in the context of this thesis. Major differences between these fields and Early Access methodologies lie in their structural frameworks and operational dynamics. For instance, crowdfunding and crowdsourcing primarily focus on soliciting financial support or contributions from a broad audience, whereas Early Access emphasizes active user involvement in the iterative development process. Similarly, community management strategies, although applicable to Early Access, may differ in scope and implementation due to the unique challenges posed by ongoing product iteration and user feedback integration.

Open innovation principles, while relevant to Early Access, may encounter barriers related to intellectual property rights and project governance that are outside from the scope of the Early Access context. Additionally, agile development methodologies may require adaptation to accommodate the broader scope of user engagement and product iteration inherent in Early Access initiatives. Furthermore, while co-development emphasizes collaborative partnerships, Early Access projects often involve a broader user community, necessitating tailored strategies for managing diverse stakeholder interests and contributions.

2.6.1. Proposed Factors Based on Literature

Recognizing these differences between fields and Early Access methodologies is essential for informing effective strategies and frameworks that address the unique challenges and opportunities presented by Early Access initiatives, thereby advancing theory and practice in both academia and industry.
Based on the literature study conducted, we assume four critical factors that need to be addressed in some aspect for any early release software development project, including game development, to successfully proceed through the process.

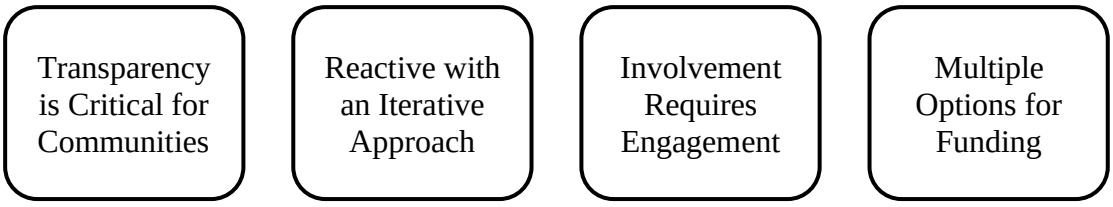


Figure 2.6-1: Core Pillars based on Literature

3. METHODOLOGY

With the previously mentioned scarcity of existing theory around the Early Access model and its impact on processes and frameworks, our research took the form of an exploratory study to establish hypotheses around the core elements that are intrinsic to the Early Access model. The approach to go with exploratory studies has been often adopted as research methodology when either it is hard to get solid baseline facts since existing facts are fragmented or contradictory, or there is a general lack of knowledge about the subject (Colin Elman et al., 2020). To ensure that the findings of our study satisfied the rigour associated with scientific studies, we referred to existing methods for generating novel theories about the Early Access model through inductive reasoning.

3.1. Data Collection Process

As with any study, our exploratory endeavours started with the process of collecting data that could be used for further analysis. Given the relative scarcity of existing academic knowledge on the matter, the choice to proceed by conducting a series of semi-structured interviews with industry professionals was straightforward as it facilitated a balance between predetermined questions and the flexibility to delve into emerging themes. that were created with initial insights collected from our literature study.

In preparation for these interviews, we referred to our introductory investigation and our literature study to shortlist the broader topics that we needed to cover with our interviews. This was then used to prepare an overall interview guide that could be referred to during the interview process (Interview Questions 8.2). To ensure that there was limited bias within the interviews we conducted, the questions were intentionally designed to be relatively open, with the broader topics to cover being the only restrictive element (Adams, 2015).

With the initial preparations completed, the next step was to gather our sample set of industry experts that would partake in said interviews. For this process, we initially contacted game development clusters and associations across Europe and the United States as initiators for introductory conversations with developers. Along with this, we expanded our efforts to include direct contact with game development studios globally. Due to a scarcity of responses to these cold reach-outs, we instead opted for a more “applied approach” by leveraging specific knowledge about communication channels within the gaming industry. Since a major component of our study was to investigate the community element, we began connecting directly with the gaming communities that we were looking to investigate. With insight that Discord was the platform of choice used for community interaction in the gaming industry, we joined numerous official Discord servers of studios and games that used Early Access as a release method and reached out directly to server administrators. With their assistance, we were able to initiate contact and schedule interviews directly with the full complement of community managers, developers, publishers, and complete executive teams as well. A summary of the resulting pool of candidates was compiled and can be found in the following section (3.1.1), along with a more detailed list of demographics that has been included in the appendix (8.3).

3.1.1. Demographic Summary of Interview Candidates

Count	**15** Candidates / **13** Gaming Studios / **2** Publishers / **18** Projects
Titles	Community Manager / Co-Founder / Developer / Publishing Partner
Job Functions	Marketing / Communications & PR / Art & Design / Development
Locations	**7** Countries
Studio Scale	Independent Studios / Subdivisions of Large Studios / AAA Studios

Figure 3.1-1: Summary of Candidate Demographics

3.2. Analysis Method

As our study primarily relied on qualitative data as an input, we were able to resort to the proven GIOIA method as the core basis for our data analysis efforts (Gioia, Corley, & Hamilton, 2013). This approach with the GIOIA method was also selected due to the provision of a relatively flexible framework that allowed for in-depth exploration of participants' experiences, perspectives, and insights. Following data collection, the process and framework outlined within the GIOIA method (Magnani & Gioia, 2023) was carried out for the analysis of the data. The initial step of this process involved reviewing the transcripts of the interviews to identify primary concepts highlighted by the respondents.

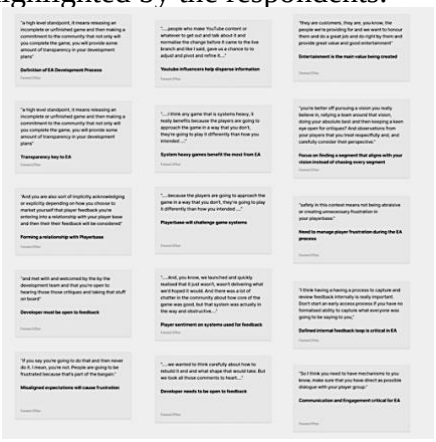


Figure 3.2-1: Sample of Initial Concepts from Interviews

The generated list of concepts was then processed and classified based on common elements and overarching subjects to highlight the relevant themes that were emerging from the collected information.

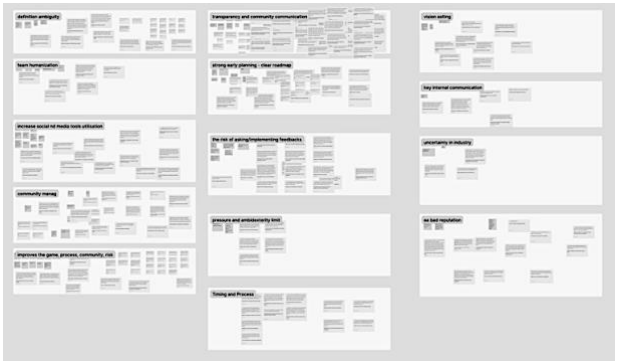


Figure 3.2-2: Concepts Grouped into Emerging Themes

Once the complete list of second-order themes had been identified, they were used as the guiding reference for establishing core aggregate dimensions that correspond to the core elements of Early Access. To summarise the overall process so far, Figure 3.2-3 represents an example of the Gioia methodology being applied wherein the different concepts collected from

interviews are clustered into broader second order themes, which are then further classified into the resulting aggregate dimensions.

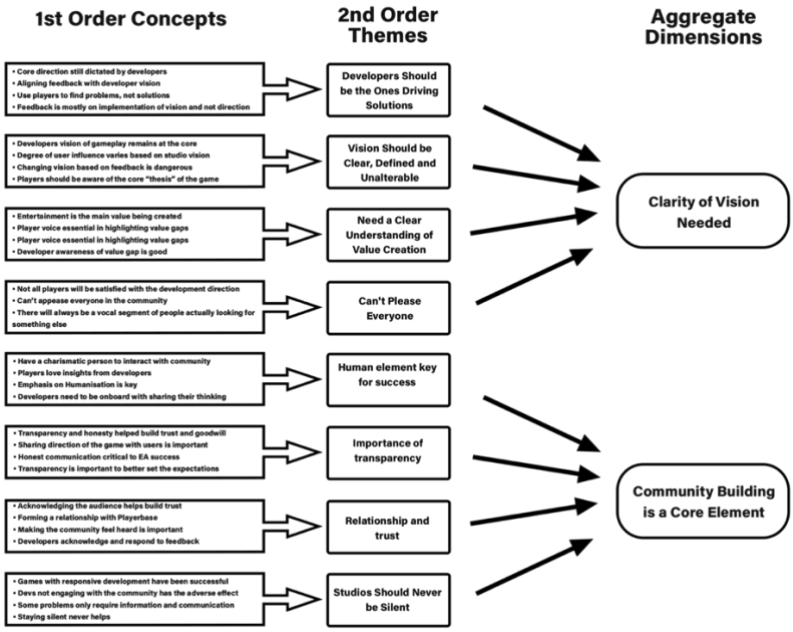


Figure 3.2-3: Overview of GIOIA Method

The analysis process then concluded with an effort to expand each aggregate dimension to further understand the different factors present within them and their relation to the overall likelihood of success for any Early Access release.

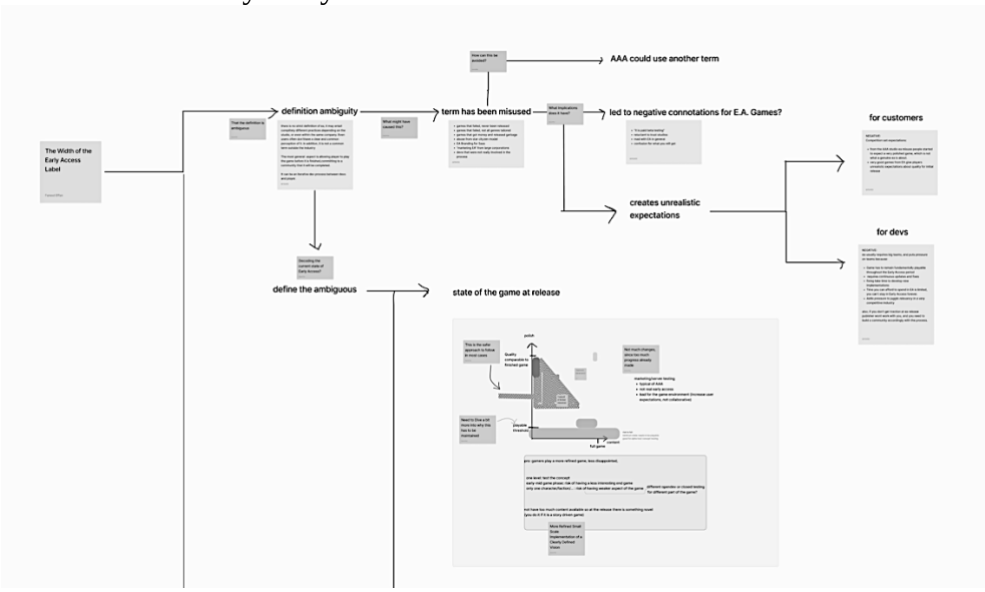


Figure 3.2-4: Example of Success Factor Analysis

4. RESULTS AND ANALYSIS

4.1. Introduction to Results

Following the process and methods outlined in the previous chapter, our study arrived at six core dimensions that could be considered key to navigating a successful game release using the Early Access model. Each of the resulting dimensions was then expanded with more details and insights to provide deeper insights and guidelines. The following sections in this chapter outline some of these findings, with relevant literature being referenced for comparison and clarity.

4.2. Early Access - An Expansive Label

With the growing number of Early Access (EA) releases, the practices entailed under the "Early Access" banner have expanded and diversified significantly, with the current term referring to a wide range of processes and methods. Conversely, the expanding width of this label has also been reflected in a growing ambiguity of its definition.
In general, "Early Access" as a term defines a game that is released to the public in a pre-1.0 state, but the more specific the formalization becomes, the more fragmented it appears, leading to disparate interpretations depending on the studio, game, and sometimes even within the same team. Further amplifying the confusion, even users often don't have a clear and common perception of what the label may entail, and the term itself is infrequent outside of the gaming industry.
The most general aspect is allowing players to play the game before it is finished, committing to a community that it will be completed. In effect, this becomes a promise made between the studio and the player base wherein they are investing in the creative vision conveyed for the game, beginning an iterative process between them and the developers with the intent of co-developing the product.

4.2.1. Historical Challenges with Early Access

One of the root causes highlighted in our findings that contributed majorly to the increasing vagueness and ambiguity of the term, even when it is so prevalent within the industry, is the historical misuse of the term over the years. Numerous projects never saw the light, even after years of collaborative iterations with the community and funding, some of which have even been suspected just to be a way of gaining income and releasing underdeveloped games. This is the same aspect of the Early Access process that was highlighted in a prior study conducted in 2021 (Mullen, 2021). Our findings corroborate this historic infamy associated with the term with two major detractors being highlighted by various respondents.
The first was the case of Star Citizen, a large-scale scale ambitious space exploration game, that has been labelled as EA for more than a decade and has raised upwards of 600 million USD while still lacking a concrete release date for its full release. Alongside this Larger AAA studios have been singled out as the second major contributor, with many using the label as a glorified marketing exercise wherein they resort to pre-releasing complete games, sometimes with an additional cost, without any intentions of co-development. These two dynamics have often been regarded as an abuse of the Early Access term, serving to discredit the whole label within the industry.
These occurrences have led to relatively negative associations with the term being formed within user communities as well, with some customers referring to it as nothing more than

“paid beta testing”. This has left users reluctant when it comes to trusting studios that associate themselves with the model.

Another ramification of the historic misuse of the label by larger corporations is the impact it has had on the expectations of the player base. Our results showed that the recent usage of the term has led to the creation of unrealistic expectations for both the developers as well as the player base. This mainly refers to instances where relatively polished AAA games have been branded as Early Access for a monetary benefit, resulting in users expecting extremely optimized and faultless games from developers. From the developer's perspective, this undermines the entire ethos of the Early Access process as it is meant to allow them to get iterative feedback on incomplete products instead of being forced to compete with extreme levels of polish. This phenomenon corroborates findings from another study that investigated the impact of aesthetics and early impressions of games that had negative receptions when they were released into Early Access, showing that it has a direct impact on customer co-creation over time (Schlãœter, Waldkirch, & Auernhammer, 2021).

On the developers' side, the Early Access label also generates additional pressure, requiring large teams and extensive commitment. This is because games need to be continuously updated and fixed since they need to remain playable for the entire duration of the process, alongside taking time to develop new iterations. As a result, the time that a studio can reliably afford to spend in EA is often limited and adds additional pressure to juggle relevancy in a very competitive industry. Additionally, this emphasises the need to carefully plan community-building efforts and activities to coincide with key milestones throughout the whole process, both for an effective co-development dynamic and to get traction at the release and obtain interest from publishers.

4.2.2. Impact of Release State

Another key element that contributes to the wide diversity seen under the label is the various release states that games can adhere to when launching into Early Access. One key finding from our study is the existence of a zone wherein the resourcing trade-offs come into heavy consideration during the design and development process for any game.
This approach suggests compromising on the polish of the game past a certain degree would put it under the threshold of "playability" resulting in a high-risk area which corresponds to an increased chance of failure both due to the limited ability to test content, as well as negative player experiences. On the other end of the spectrum, there is another approach that focuses on a significantly smaller amount of highly polished content being the initial release. While this alleviates some of the risk associated with a negative first impression, our results suggested that this can have an impact on the long-term success of the game as well due to the lack of testing associated with late-game content.
The diagram below (*Fig.4*) serves to visually represent some of these approaches to balancing content and the different zones that correspond to the different release states of an Early Access game, along with some implications for where developers position their initial release. The diagonals represent different degrees of resource availability, showing how the trade-off might scale across different allocation approaches. The extremities for both approaches showcase the extremes, which would significantly increase the risk associated with any release. Along with this, any zone that corresponds to areas where the limitation of resources is no longer a concern for the scale of the game can be considered the territory of larger-scale studios with ample development resources.

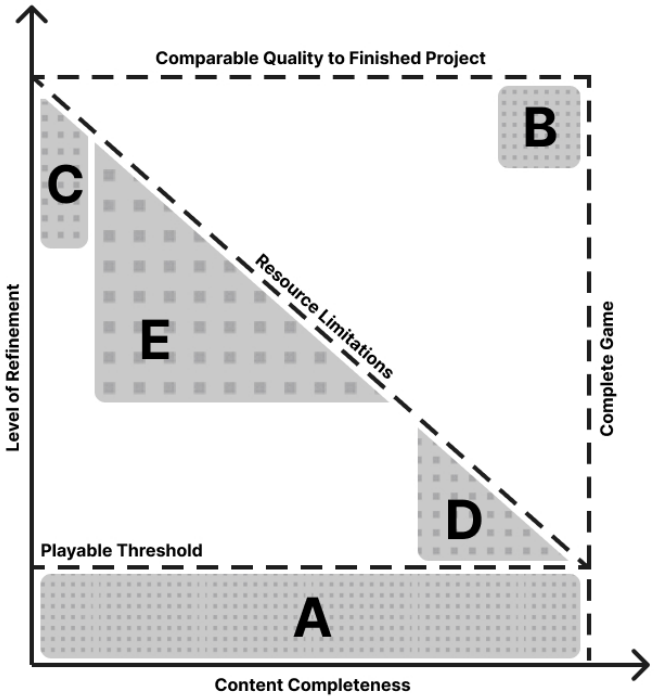


Figure 4.2-1: Visualising Trade-off Between Content & Refinement

The representative areas in the diagram reflect notable aspects and scenarios of the impact of the release state:

4.2.2.1. **Zone A –** Risk of Unplayability

While considering the amount of game content in an Early Access release, the first rule that almost every single interviewee pointed out is the need for the game to be always fundamentally *"playable"*. This zone represents a high-risk area corresponding to an increased chance of failure both due to the limited ability to test content, as well as negative player experiences. Releasing what it is often called "a broken game" is the easiest way to guarantee failure, as even pre-release content often has high player expectations. This approach can immensely damage the reputation of the whole project. For that reason, any game release needs to be "above" the playability threshold when it is released.

4.2.2.2. **Zone B –** Misuse of EA Label by AAA Studios

As explained in previous chapters, large studios that aren't constrained to such resource limitations often misuse the Early Access label for marketing purposes, delivering earlier access to a relatively polished complete product for a higher price.

4.2.2.3. **Zone C –** Extreme focus on Quality

The area reflects an approach that focuses on a small amount of highly polished content being the initial release. While this alleviates some of the risk associated with a negative first impression, our results suggested that this can have an impact on the long-term success of the game as well due to the lack of testing associated with late-game content. This approach serves as a good methodology for games where core game mechanics are very central, and often are not story based. An example of this case is when an Early Access game gives the possibility to the gamers to use only one or some of the available characters, factions or in some of the levels, to test the main game dynamics.

4.2.2.4. **Zone D –** Prioritising Content Quantity

The opposing extreme to the previous approach, this method instead focuses on the initial release being as close to content complete as possible, regardless of the compromises made in terms of quality and polish. This scenario can be quite risky as it places the game near the *"unplayability threshold"* mentioned earlier.

4.2.2.5. **Zone E –** Balanced Content Quality and Quantity

After considering all the other extremes on the diagram, this one zone is where a studio can limit its risk of failure, as well as increasing the chances of successfully navigating the Early Access release.

4.2.3. Multiple Phases of Availability

The final cause that contributes to the splintered definition of Early Access that was highlighted in our findings was the existence of multiple phases of availability throughout the design and development process. These phases can be added or removed depending on the approach taken by any studio and depends on how they intent leverage the different phases to develop and iterate towards their final release. Alongside this, one consequence that was commonly highlighted in our results was the correlation between the scale of access and the amount of actionable feedback that can be collected by the studio.

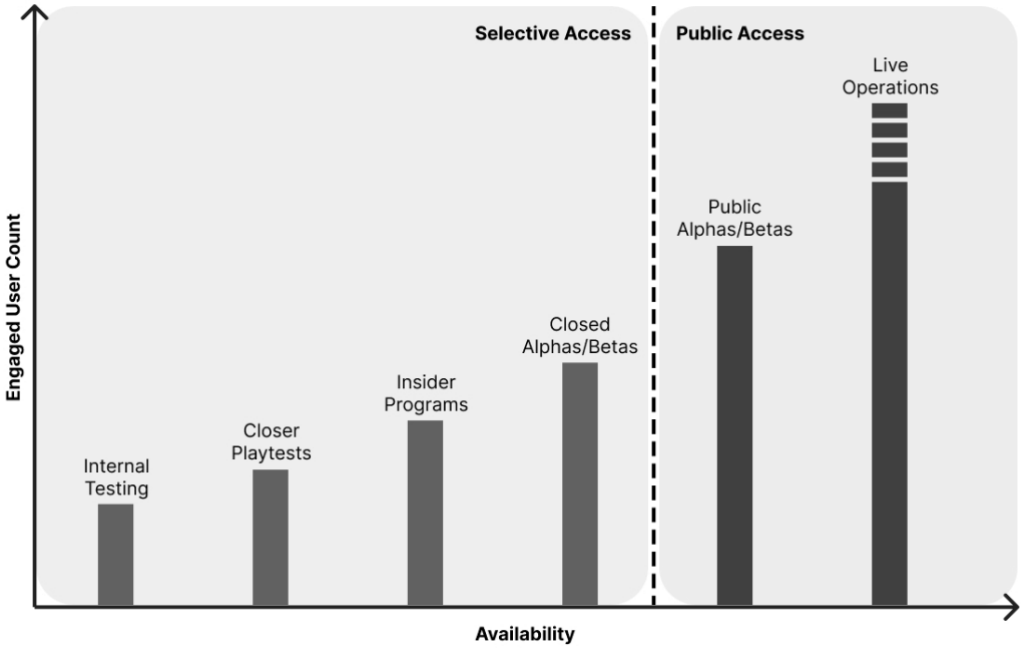


Figure 4.2-2: Common Phases of Selective and Public Access

The diagram above visualises some of these different phases that were highlighted in our findings transitioning between selective access, where only closed segments of users are given access, all the way through to the live operations stage wherein the game is widely available to the public. The scaling count of engaged users seen as the game development progresses through the phases is also indicative of the level of feedback that the studio can expect, with selective access severely limiting the amount of insight and feedback that can be collected and acted upon.

Enlarging the scope of users that the game is available to, the amount of feedback that can be collected increases substantially. The first example of this often seen is the open alpha / beta stage, wherein a test build can be made available to the public. Expanding on this even further, the live operations segment refers to the approach that treats a game as a complete live service product and continually delivers new features, updates, promotions, in-game events, and improvements to the launched game to improve the experience for the player community.

The chart highlights how public access, even if requiring a product closer to the 1.0 state and substantially higher efforts to be set up and managed, gives an extremely bigger amount of actionable feedback, impossible to obtain with closed testing in a reasonable amount of time.

4.3. Clarity of Vision and Its Importance

Among the most critical elements to the success of any firm is the overall vision that governs both their operations as well the products they work to develop. While this is true for all businesses to some degree (Walt, Kroon, & Fourie, 2004), our discussions with studios showcased that it has a much larger impact when it comes to game development.
Many of our respondents, which included various developers, community managers and founders, stressed the importance of the overall vision and strategy set by the studio. This was frequently referred to by our interviewees as a core factor that played a key role in the success of any game that undergoes the Early Access development and launch process. However, one thing that did vary widely between the different interviews was the depth of the strategy and the aspects within the studio that it influenced.
These discussions could be classified into two core groups, with the first focusing more on the internal processes within the company, which showcased the link between strategy and aspects like the creative direction of the game, the development approach that was chosen, as well as the overall production planning from ideation to release. Conversely, the other perspective mainly linked the strategy to outward-facing aspects of game development, focusing more on factors such as communication, messaging and branding, as well as the understanding of the value proposition for the game.

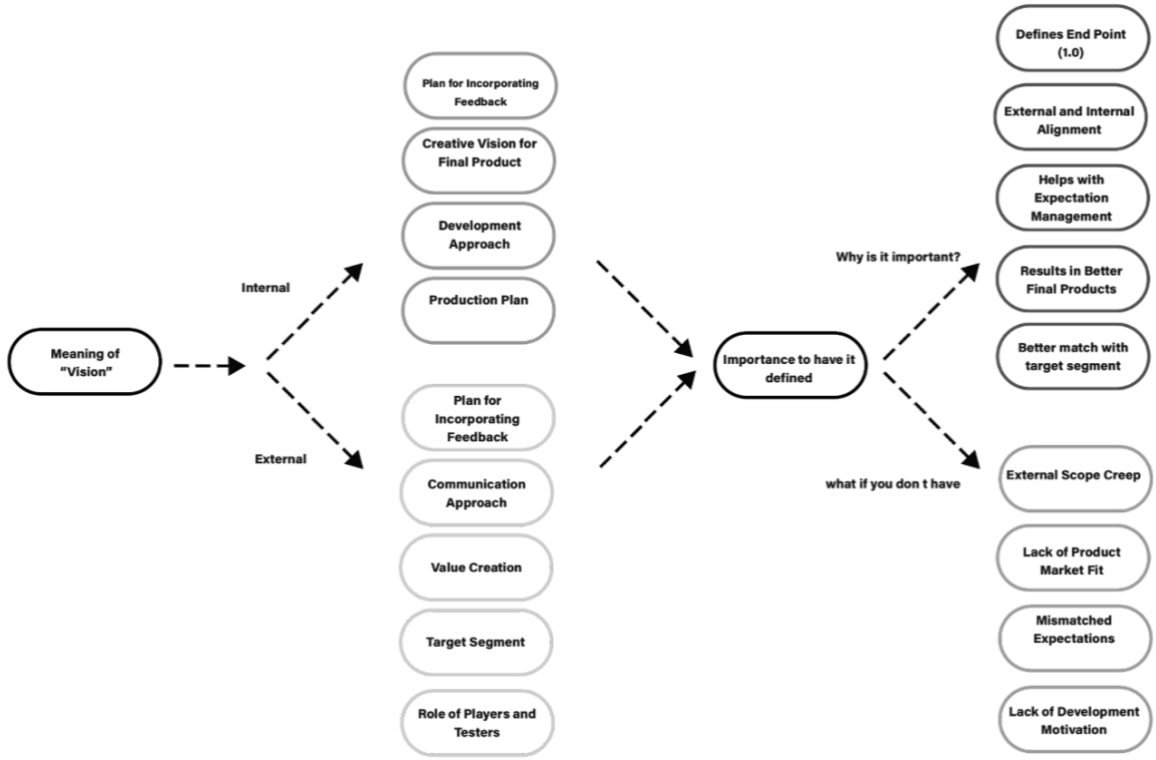


Figure 4.3-1:Internal and External Impact of Product Vision

Building on this further, two core links were commonly made between the overall vision and the role it commonly plays in ensuring alignment between the external and internal aspects, as well as the core influence it has on the feedback process. While the feedback loop is another finding that is covered in detail in a later segment, the link between internal and external factors was emphasised as a crucial element as to why a well-defined strategy was needed for any game to successfully navigate Early Access. Further cementing this, another core theme that was present among all interviews, was an emphasis that the final decisions relating to the vision and creative direction, as well as whatever is influenced by them, should remain within the studio's control and that there should be a clear and defined strategy in place before heading into the design and development process.

When faced with questions on the possible ramifications of lacking such a defined strategy, many responses shared common elements that highlighted issues that pertained to fundamental issues such as lacking a core product market fit or facing severe external scope creep. All of these can compound significantly when combined with the mismatched expectations these issues can cause with the player base, as the gap between the product that can be delivered by the studio and the value players are looking for grows.

The insights from our study around the depth of impact that the overall vision can have within game development studios also match with the findings from a prior study, where one such studio's approach to developing a morally challenging game was investigated. Their study had similar findings that showcased the degree to which the overall vision deeply influenced key decisions throughout the game development process (Smale, Kors, & Sandovar, 2019).

4.4. Multiple Variations of Early Access

The multiple variations of Early Access methodologies reflect the numerous development opportunities that these practices provide. These potentials fall into three main dimensions: risk management, founding, and game improvement.

4.4.1 Risk Management

The Early Access game release methodology significantly mitigates the financial risks associated with game development through several key mechanisms. Early Access enables developers to gauge market interest and assess the viability of their game concept before investing substantial resources into its completion, it allows the studio to “fail early” if the game fails. By obtaining direct feedback from players, developers can identify potential issues early on and make informed decisions to address them, thereby minimizing the risk of investing in features or content that may not resonate with the target audience. Furthermore, the iterative nature of Early Access allows developers to prioritize development efforts based on player demand and engagement, ensuring that resources are allocated efficiently to areas that will yield the greatest return on investment. Overall, the Early Access game release methodology serves as a strategic tool for managing financial risk in game development by providing a sustainable revenue stream, facilitating market validation, and enabling iterative refinement based on player feedback.

4.4.2 Funding

Early Access in video games serves as a vital source of seed funding, particularly for small studios that may lack the financial resources to complete game development without it. For these developers, the upfront revenue generated from Early Access purchases can be instrumental in covering critical expenses such as software licenses, equipment, and salaries. Without access to traditional funding sources or established financial backing, Early Access provides a lifeline, enabling small studios to kickstart their projects and sustain development efforts. This initial injection of funds not only helps cover immediate costs but also instils confidence in the project's viability, attracting additional investors or partners down the line. In essence, Early Access serves as a democratizing force in the game industry, levelling the playing field and empowering independent developers to bring their creative visions to life, despite limited financial means.

4.4.3 Game improvement

The Early Access game release methodology serves as a dynamic platform for fostering continuous improvement and innovation within the gaming industry. With an initial Early Access release, developers gain the opportunity to not only sustain their development efforts but also expand the scope and targets for their projects. This extended development period allows for a more comprehensive exploration of various game aspects, including mechanics, features, and content. Developers can experiment with new ideas, iterate on existing systems, and implement player-driven suggestions to enhance the overall gaming experience. Furthermore, the iterative nature of Early Access facilitates a continuous feedback loop between developers and players. This collaborative process ensures that the game evolves in response to player preferences, community feedback, and emerging trends in the gaming landscape. By actively engaging with their player base, developers can identify and address issues, refine gameplay mechanics, and optimise balance to create a more immersive and enjoyable gaming experience. This approach is particularly beneficial for system-heavy games, which often require extensive testing and fine-tuning to achieve optimal gameplay balance and depth. Through Early Access, developers can leverage the diverse approaches and perspectives offered by players to refine their game systems, resulting in a more compelling and rewarding

experience for all involved. Finally, in general, numerous practitioners believe that games are generally more balanced and enjoyable when going through the Early Access.
Notably, these different benefits from a fruitful EA development are strictly connected to the attention to the most critical phase of the process: the initial launch. The Early Access launch phase is undoubtedly a pivotal moment for game studios, as it sets the stage for a multitude of benefits that can significantly impact the success and longevity of a game. This phase is particularly crucial because it marks the transition from internal development to engaging with the wider gaming community. With an internal feedback loop already defined and a community built according to the roadmap, a well-approached initial launch serves as the cornerstone for future success. Firstly, it provides the necessary traction to carry out the Early Access phase. This initial burst of momentum is vital for attracting the attention of both players and industry influencers, creating buzz and anticipation around the game's development journey. It's during this phase that developers have the opportunity to showcase their vision, garnering support from early adopters who are eager to be part of the game's evolution. Secondly, an impactful launch generates attention from customers, drawing them in with promises of involvement in shaping the game's future. Through strategic marketing efforts and community engagement, studios can leverage the excitement surrounding the Early Access launch to attract a wider audience and increase visibility within the gaming community. Thirdly, the Early Access launch phase serves as a catalyst for expanding the community. As players begin to experience the game firsthand, they become more invested in its development, forming connections with other players and engaging in discussions about their experiences and suggestions for improvement. This sense of community not only fosters a supportive environment for developers but also encourages ongoing player participation and feedback throughout the Early Access phase. Additionally, a strong launch contributes to increased player retention. Early adopters who feel a sense of ownership and investment in the game are more likely to continue playing and supporting its development over time. By fostering a positive and engaging experience from the outset, studios can cultivate a loyal player base that remains dedicated to the game throughout its development journey and beyond. Ultimately, the initial launch phase represents the period with the highest relevancy, setting the tone for the entire Early Access journey and shaping the game's trajectory towards success.

4.5. Community Building is a Core Element

With the growing utilisation of models like Early Access, this complex relationship between studios and their invested players is something that has become even more significant as time goes on, with an increasing significance placed on bi-directional communication. From our discussions, we've learnt that this has resulted in an increasing emphasis on activities and factors that are critical to community building, as developers become more reliant on their user base to collectively test and aid their efforts in creating unique and diverse worlds.

In this case, community building refers to the strategic efforts made by such game developers, publishers, and other related stakeholders, with a pivotal role played by community managers, to cultivate and engage a dedicated and active player base. It necessitates accurate planning, with a well-defined roadmap established before the pre-launch phase. Before and during EA phases, the community collaborates to address shared challenges, express their enthusiasm for the game, and, notably, shape its evolution. On the developer's end, the selection and maintenance of community tools are crucial elements, contributing to the creation of a space for constructive discussion while mitigating toxic behaviours and discourse.

Typically, Early Adopters are the most invested, and close collaboration with these influential community members is vital for the stability of the community itself, as they can become advocates for the developers and the studio. The ultimate objective of constructing a robust community lies in enhancing the final product, by helping with the external and internal alignment and expectation management, guiding the definition of the Early Access endpoint (1.0 release), and ensuring alignment between the game's value proposition and the target segment's perception.

Furthermore, a community-managed knowledge base adds value to the game, contributing to collecting and sharing information and guides, facilitating the growth and interest of the game. One respondent phrased this in a rather concise way, highlighting that with a good relationship and foundation with the developers in place, a strong user community can even tolerate relatively broken and unpolished experiences if the trust exists in the developer to eventually realise their vision for the product. In the dynamic landscape of the gaming industry, community building serves as an anchor for sustained Early Access success, and several enablers and practices are essential for its effective implementation. First and foremost, developers must possess a clear idea or vision of the community they aspire to cultivate and articulate what they seek from it. Establishing a roadmap that aligns with the overarching goals of the game is essential, delineating a strategic path for community growth.

The initial phases of community building are particularly critical, laying the foundation for long-term engagement and satisfaction. Strong communication is an indispensable practice: showcasing developers' active involvement in the game's evolution fosters a sense of transparency and connection. Timely updates, developer insights, and responsiveness to player feedback contribute to a healthy dialogue between developers and the gaming community. Additionally, empowering players to share their creations, whether through user-generated content or mods, has a profound impact on community dynamics, promoting creativity and collaboration that extend beyond the confines of the game itself. These enablers and practices collectively form the backbone of a flourishing gaming community.

It is also important to be aware of the existence of "blockers" and toxicity, which pose significant challenges. Communities can also simply be groups of individuals sharing a common interest in a particular game, without any commitment to give actionable and valuable feedback to the developer teams. Hence, the unregulated nature of these collectives can sometimes lead to toxicity. Developers, facing the potential backlash and unwarranted criticism, may opt for a more private stance, limiting their public engagement.
With the insights collected from our discussions with key industry roles in the game development process, we formulated a set of fundamental pillars that could be considered core contributing factors for building strong communities.

4.5.1. Maintaining Transparent Communications

One key topic that was continually brought up when covering the topic of communities and the factors that impact their development growth was the substantial importance of maintaining a high degree of transparency in any discourse in the community. Early Access releases are especially dependent on this, where transparent communications between the studio and the player base play a key role towards establishing trust and goodwill, a foundation that must be upheld consistently throughout the development process.
This involves honesty in sharing the game's creative and development direction with users, along with providing key insights into the development progress and decisions made throughout each stage. This effort becomes critical during the initial Early Access release, however maintaining transparent communications throughout the development process helps set realistic expectations, which play a core role in creating and sustaining engaged communities that are invested in the success of the game.

4.5.2. Humanisation of Development Teams

Another key element that surfaced repeatedly during our study was the impact studios observed when focusing on efforts towards humanising the development teams by encouraging and enabling developers to directly engage and partake in the player communities. All the studios that had undertaken efforts to support and encourage developers to become involved in the community discourse suggested that it had a significant contribution towards building a strong and involved player base, and simultaneously on the success of their game.
Many respondents also highlighted the impact of leveraging charismatic individuals as spokespersons with an ability to paint vivid pictures of the product vision and to take the community on a collective journey with the developers throughout the Early Access process. This increasing emphasis on the human side of the teams behind the game can be an instrumental factor in fostering a deeper understanding between the developers and the community.
This approach does have its caveats, however, with a greater responsibility being placed on the teams behind the games to be more vulnerable in putting themselves in front of the community. Developers must be willing to share their thoughts and rationale, actively engaging with the community from the outset. Direct interactions between developers and the player base, for example, on platforms like Discord, is a key component towards this effort. This growing link means that studios must also acknowledge that developers are as integral to the community as communication managers in reinforcing the collaborative nature of the overall gaming experience as well as the Early Access process.
These findings coincide with the conclusions from a prior study that investigated the activities developers undertake and their impact on the formation of communities in large-scale online massively multiplayer online games. One of the core results from that study puts a key emphasis on direct developer engagement as a major catalyst in the formation of strong online communities (Ruggles, Wadley, & Gibbs, 2015).

4.5.3. Importance of Consistent Communication

Failing to communicate changes to the community poses inherent risks. Successful games released with EA often attribute their community-building achievements to the responsiveness of their teams. At the same time, a lack of communication has frequently led to negative responses from the community. The importance of communication is particularly crucial during times of crisis, when, for instance, the game is facing issues and playability is at risk.
Responding promptly to the community, explaining how teams are working towards fixing issues, and having the users informed and involved helps build and maintain engagement, lower discontent, and preserve a positive reputation. On the other hand, remaining silent in any situation is never beneficial, and can lead a community to collapse, becoming toxic and damaging player retention. It is crucial to recognize that certain issues can be effectively addressed through the provision of information and open communication.
Finally, building a robust relationship between developers and the community is central. On the developer side, a willingness to actively engage with the community is foundational. Both proactiveness and reactiveness to player feedback are crucial, demonstrating a commitment to continuous improvement to the project. Acknowledging the audience's presence not only cultivates a sense of trust but also generates a more engaged and honest community. Leveraging the trust established through past studio releases, developers recognize that reputation often compensates for any initial game incompleteness, although it's not deemed critical for the success of Early Access.
On the user side, providing players with choices is important, and content users are more likely to trust and accept changes. Direct communication with players facilitates discussions about features, making the community feel like an integral part of the development process. Ensuring that the community feels heard is as important as acknowledging their frustrations, these principles lead to a genuine connection between consumers and developers. An ideal community lies in the middle road where users are open to criticism but in a constructive way.

4.6. Expanded Approach to Feedback

The overall approach to handling feedback was also an area which stood out and was frequently linked with the importance of the overall vision that was laid out by the studio. Participants highlighted that the overall approach to feedback evolved drastically as Early Access gained more traction and an increased focus was placed on iterative and collaborative development within the industry. A common focus point in the responses was that the collection and proper implementation of feedback was a critical aspect of Early Access development and that studios should be sufficiently invested in that process if they are looking at Early Access as a development alternative.
Conventionally, the feedback chain was comparatively more unidirectional and was often preserved and documented as learnings for future products and releases. This was compounded by the fact that limitations in processes meant that often developers were unable to implement feedback in existing products in time.

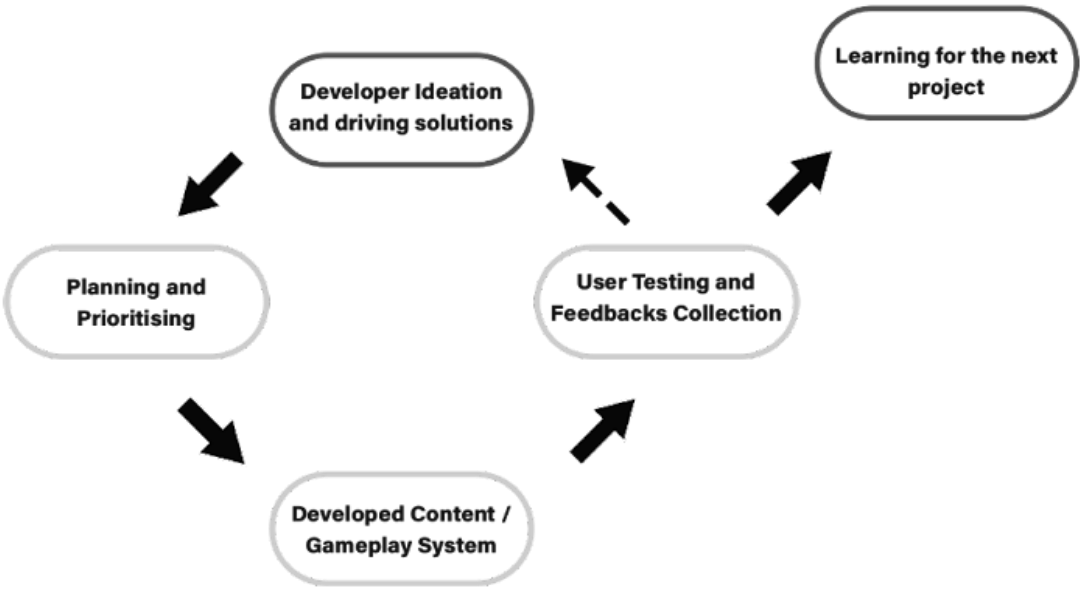


Figure 4.6-1: Conventional Feedback Loop

Within the topic of feedback and information flow between the development teams and the player base, three key themes emerged frequently as core process evolutions that have evolved to adapt to the collaborative development requirements seen in the Early Access process.

4.6.1. Evolved Feedback Loop

Compared to this, the feedback loop for Early Access games has become considerably more complex, with a particular emphasis on bilateral communication between the users and the development teams. This focus on information dispersion and the important role it plays in an Early Access development cycle was best showcased when a trend emerged that many common issues can be solved simply by sharing information more frequently with end users.

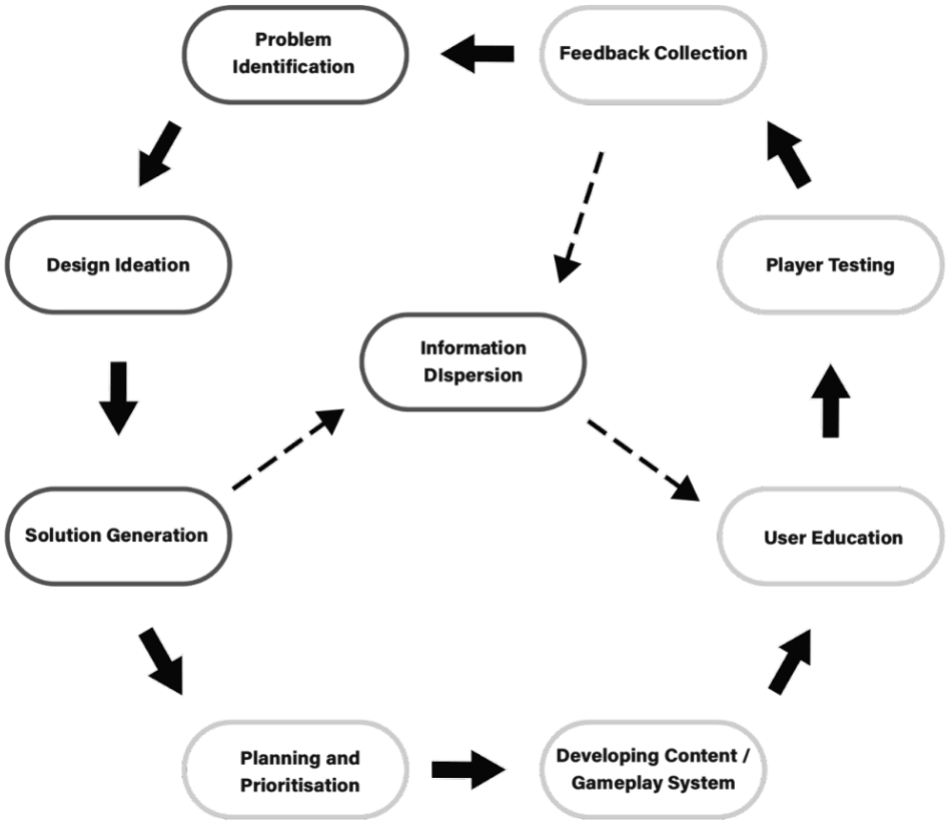


Figure 4.6-2: Evolved Feedback Loop for Early Access

4.6.2. Methods for Information Dispersion

Based on an approach that placed information dispersion at the centre of the feedback loop, there were also some notable insights shared about the different options available for managing information dispersion and communication with the user base. Along with this, respondents also outlined the important role this process plays in the formation of engaging communities with how it enables developers to showcase more of the human factors that go into creating these worlds.

This is something that should not be taken lightly, as misinformation is also something that must be factored in. If studios aren't proactive about managing the details being discussed and shared, it can often lead to an increase in user frustration as they are left with many different sources, and sometimes conflicting, sources of information.

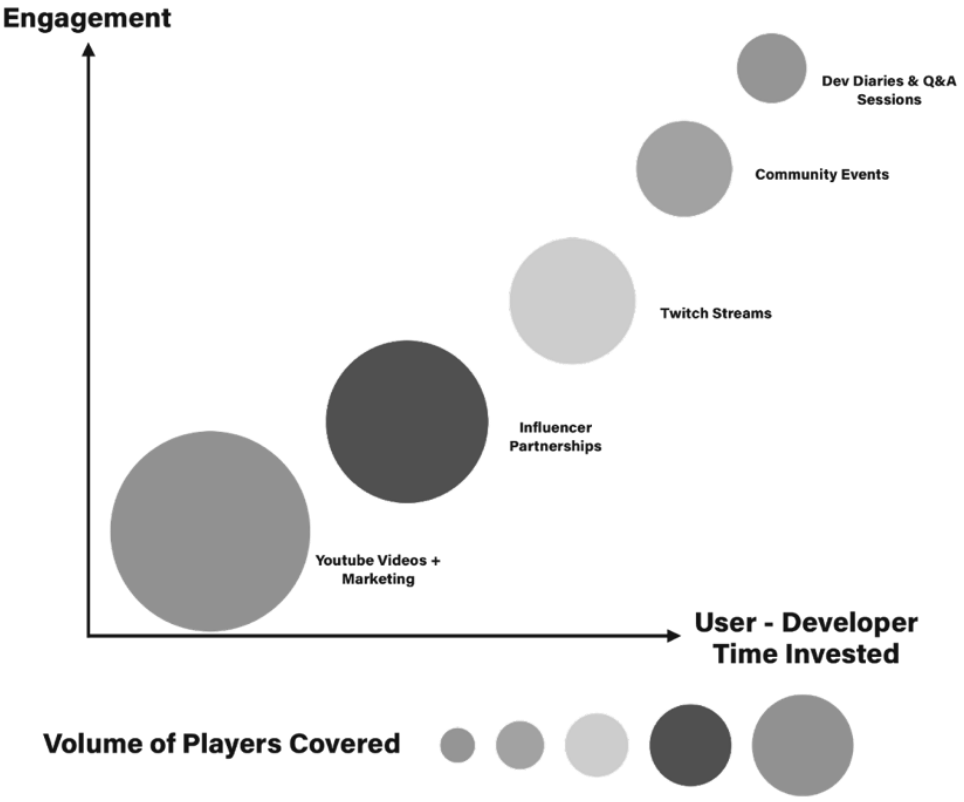


Figure 4.6-3: Methods for Information Dispersion

4.6.3. Feedback Collection Methods

Along with the overall importance of improved communications between studios and their users, the results also placed a particular emphasis on how methods for collecting feedback have evolved, with both active and passive sources of feedback being employed by developers in modern titles.

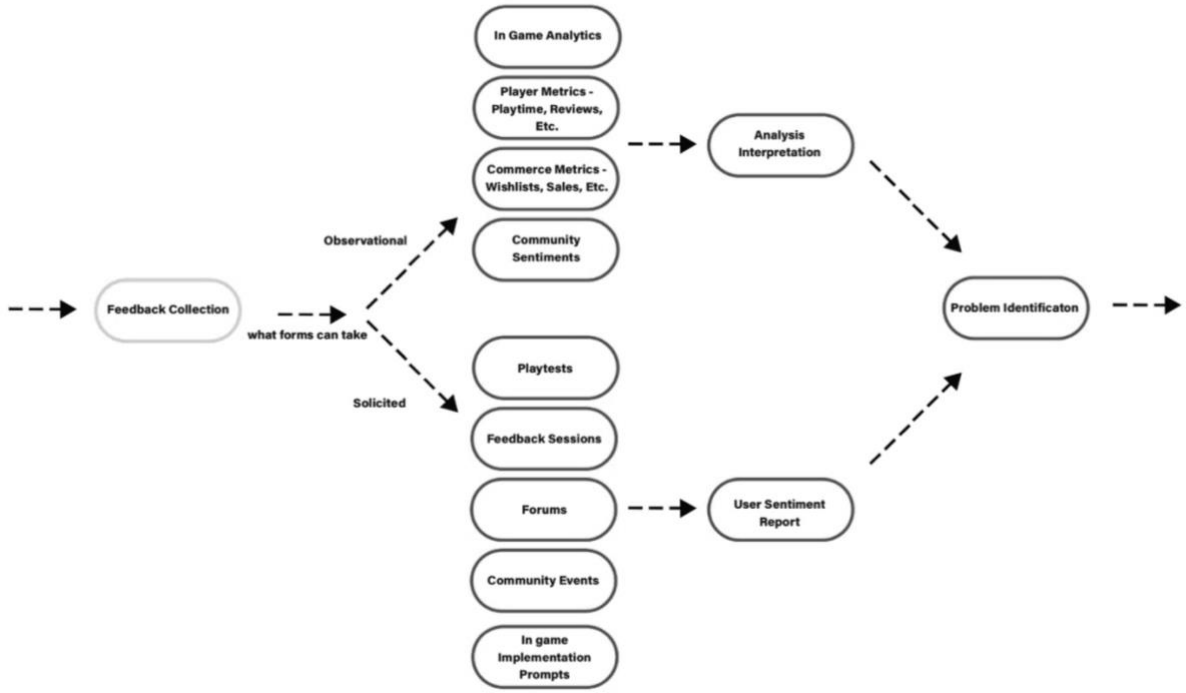


Figure 4.6-4: Methods for Collecting Feedback and Insights

4.7. Evolution of Processes and Roles Due to Early Access

Keeping in view the insights and findings highlighted in the previous sections, it also showcases the evolution of the game development process and how the industry has evolved with the increasing popularity of Early Access. Across the interviews we conducted, this was also an aspect that was discussed frequently, with most respondents sharing insights around the changing needs in the game development industry with higher expectations around quality as well as the additional challenges that are brought about by an iterative development model such as Early Access.

One such challenge that was frequently brought up was the additional effort required for communications in Early Access. Given the critical part that the entire communication loop plays in the process, studios quickly began to understand that the dedicated role was required to meet the demands of maintaining these lines of connection with the community, both for feedback collection as well as managing information dispersion. This need for a dedicated specialist in communications evolved into the role we see community managers playing today, acting to bridge the gap between users and the development team.

Another area that has undergone a noticeable evolution due to the requirements of Early Access is the testing and feedback loop. With the increasing popularity of the iterative development model seen within Early Access came a need to be able to collect and consolidate feedback from multiple sources in an efficient manner. To adapt to this, specialised tools have been developed and becoming more prevalent within the industry, enabling community managers to collect feedback and user insights from multiple sources. Two of the most insightful examples from our interviews were the use of in-game analytics to collect insight about user behaviours in the game, and the creation and use of a "community sentiment report" that enabled community managers to streamline the process of collecting and sharing user sentiments with the design and development teams.

On the topic of testing and feedback, the roles and expected contributions from different people involved in the game development process have also seen a significant change. Many of our respondents were eager to highlight the impact that user testing with incomplete builds has had on the process, as it allows for a significantly greater scale of testing and validation to be carried out than a dedicated internal testing team within the same period.

This shift towards greater user involvement in the development process has also been welcomed by the industry, particularly for games which are designed around heavily interdependent systems. Within this new role, early players have often shown a very high level of involvement in the development, while also allowing for a much wider variety of playstyles that can challenge existing systems. This transfer of testing towards the users also means that the role of Quality Assurance has evolved more towards being able to reliably replicate issues instead of carrying the burden of trying to uncover them as well.

When discussing how roles and methods have evolved to become more tailored to Early Access, it's also pertinent to discuss the impact such a shift has had on the development processes that are common within the industry. With the additional requirements seen around being able to iteratively develop and test new builds with users, studios have made significant strides in evolving their internal processes and tools to cope with these additional needs.

4.8. Proposed Updated Model

After processing the collected data through the described methods, our study resulted in a total of six aggregate dimensions that correspond to intrinsic elements of the Early Access model. These aggregate dimensions can also further be classified with a set hierarchy based on the level within the industry that they correspond to. The chart below serves to showcase this hierarchical classification, while the subsequent sections serve to expand on each of the dimensions in more detail. The order that we are presenting for the dimensions reflects their hierarchical position, based on their importance on a conceptual and practical level.

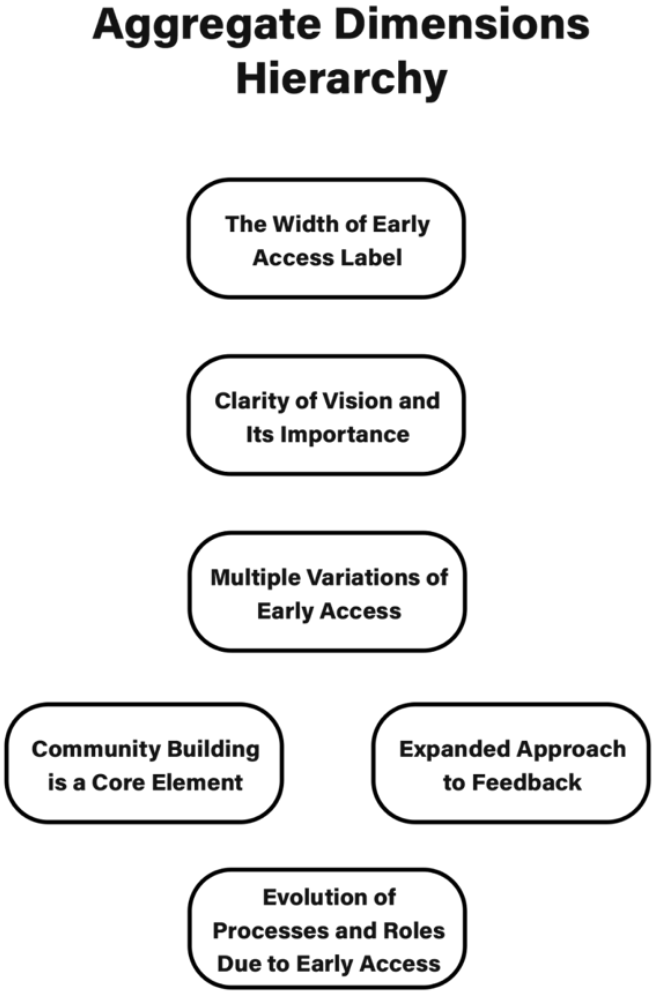


Figure 4.8-1: Proposed Updated Framework

5. DISCUSSIONS & CONCLUSIONS

With the resulting model from our qualitative study, along with our initial findings from literature pertaining to the broader software development industry, we can draw some final conclusions around the core elements within Early Access and the impact that they have on the studios efforts as they proceed to leverage the Early Access model for their products.
Expanding on this further, a direct comparison between both models provides interesting insights on the relation between the Early Access model and the methods conventionally used in modern software development efforts.

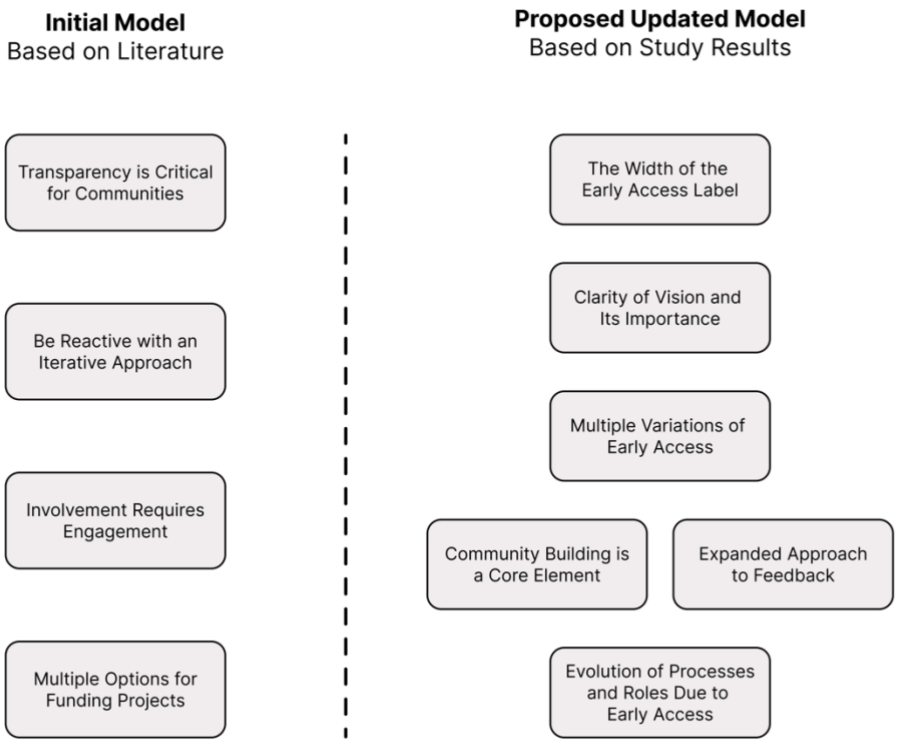


Figure 4.8-2: Frameworks Comparison

In this comparison, we can see that both approaches follow a similar trend, particularly in terms of the core focus areas that they prioritise, with a heavy emphasis being placed on elements like keeping a user centric approach throughout the entire design and development process. The continual focus on close engagement with the user was a prominent theme in the literature reviewed around software development and was referred to substantially in our qualitative study as well.
As a qualitative validation of our findings, several opinions and insights from the interviews had commonalities with the theory we researched. Hence, adjacent industries and similar processes such as open collaboration environments or open-source software development, give us a critical understanding of community management aspects. In all these approaches, community involvement is paramount, with external contributors playing a vital role in shaping the final product. Access to intellectual property is democratically granted, whether it be

through sharing ideas in open innovation, contributing to open-source software, or participating in Early Access game releases. Collaborative co-creation is a prevailing theme, as external partners, community contributors, or Early Access players actively influence and refine the result. Across these diverse domains, embracing openness proves to be a powerful strategy, fostering innovation, accelerating development, and creating a sense of shared ownership and engagement among participants. As highlighted in our literature review, when it comes to managing communities, supporting dialogue and designing iterative processes, several key principles of these collaborative environments are quite consistent also in the context of Early Access.

As with all services and products, both physical and digital, the satisfaction of the end customer must be considered in all activities and processes. The importance of the customers, their loyalty to, and relationship with any company is something that has widely been explored in various studies in the past (Arslan, 2020). The same also holds within the gaming industry, where communities that form around any game can often play a pivotal role in whether the game is successful or a commercial failure.

The role transparency plays in any community's perceived value of a product, in this case, the games created by the studio, is also an interesting reinforcement of this finding. Various studies have investigated the topics of both product and operational transparency across multiple industries, with similar findings that suggest that there is a relationship between loyalty, satisfaction, perceived value, and transparency (Buell, 2019) (Stevens, Brian, Breazeale, & Jones, 2018).). It is relevant to mention that, in line with the Transparency component in the DART model (Iglesias et al., 2013b) cited in our literature review, (Nambisan & Nambisan, 2008) unveils three different degrees of transparency: role transparency since the consumers have certain perceptions about the role they occupy within the virtual environment. Process transparency pertains to the clarity of innovation procedures, encompassing the chronological sequence, the individuals involved, and the connections to other facets of the firm's processes. Outcome transparency is keeping the consumers notified about what's happening to the ideas they participated with (Cornelio & Cruz, 2014).

Among the insights shared by our respondents, some of the most impactful examples were the evolving shift towards more nimble ways of working with development and delivery. Many studios have started to implement forms of tiered development branches that integrate with continuous integration and continuous delivery (CI/CD) pipelines to enable faster deployments that can be tested by the players to enable more frequent iterations and improvements.

This is an interesting development from the process perspective, as it results in parallels to the Agile development methods that focus on iterative development and testing combined with continuous feedback implementation, building further on the initial findings that purveyed the usage of modified agile practices within the gaming industry (McKenzie, Morales-Trujillo, Lukosch, & Hoermann, 2021).

While the usage of branched development is slightly different within the gaming industry, another parallel does exist when discussing the aspect of version management with multiple development branches being used. Our insights showed that an abundance of studios have started operating with multiple test builds and experimental branches, even within the Early Access umbrella. Allowing more invested players to give early feedback on features before a significant commitment must be made to the creative direction of the game with a given system.

6. RECOMMENDATIONS AND FUTURE WORK

While the study we conducted generated some significant insights around the Early Access development model seen within the gaming industry, as with any study, there is still room for additional investigations that address the limitations of our study. Some of these recommendations are listed below, along with some suggestions for future work that could validate or expand upon our findings.

6.1.1. Recommendations

Expand based on Implications for Theory and Practice
One recommendation that would serve to address a key limitation we faced would be to repeat the same process with an expanded list of studios, while also increasing the variety of roles included for each studio.
Another aspect to consider is the rising prevalence of game subscription services like the Xbox Game Pass that play a similar role in lowering the barrier to entry into game development and minimising some of the risks seen by independent developers. Expanding the investigation to include these subscription services in some way would provide a more holistic view of the position the Early Access model has within this constantly evolving industry.
Finally, an interesting parallel that was raised during our study was the rising popularity of live service games in recent years. These products showcase some of the same elements of co-development seen within Early Access and would be quite valuable to investigate.

6.1.2. Future Work

A recommendation for a follow-up study would be to include quantitative data around key industry metrics as well as economic markers for commercial success for any game to validate the core elements highlighted in our study.
Along with this, another possibility for a follow-up study would be to investigate the relationship between these core elements of Early Access using relationship modelling tools.

7. WORKS CITED

Poor, N. (2014, December). Computer game modders' motivations and sense of community: A mixed-methods approach . *New Media & Society*, pp. 1249 - 1267.

Behrenshausen, B. G. (2013, September). The active audience, again: Player-centric game studies and the problem of binarism . *New Media & Society*, pp. 872 - 889.

Reddy, M. (2011). *API Design for C++*. Morgan Kaufmann.

Grand View Research. (2023). *Video Game Market Size, Share & Trends Analysis Report By Device (Console, Mobile, Computer), By Type (Online, Offline), By Region (Asia Pacific, North America, Europe), And Segment Forecasts, 2023 - 2030.* Grand View Research.

Landoni, P., Dell'era, C., Frattini, F., & Petruzzelli, A. M. (2020, April). Business model innovation in cultural and creative industries: Insights from three leading mobile gaming firms. *Technovation*, pp. Volumes 92–93.

Sandqvist, U. (2015, Autumn-Winter). THE GAMES THEY ARE A CHANGIN': NEW BUSINESS MODELS AND TRANSFORMATION WITHIN THE VIDEO GAME INDUSTRY. *HUMANITIES AND SOCIAL SCIENCES LATVIA*, pp. 4-20.

Lin, D., Bezemer, C.-P., & Hassan, A. E. (2017, July). An empirical study of Early Access games on the Steam platform. *Empirical Software Engineering*, pp. 771-799.

Plum, O., & Hassink, R. (2014, October). Knowledge bases, innovativeness and competitiveness in creative industries: the case of Hamburg's video game developers. *Regional Studies, Regional Science*, pp. 248-268.

Bagherzadeh, M., Markovic, S., Cheng, J., & Vanhaverbeke, W. (2019, January). How Does Outside-In Open Innovation Influence Innovation Performance? Analyzing the Mediating Roles of Knowledge Sharing and Innovation Strategy. *IEEE Transactions on Engineering Management*, pp. 740 - 753.

Mullen, M. (2021). Crash and Burn. In M. Mullen, *The Ethics of Playing, Researching, and Teaching Games in the Writing Classroom* (pp. 23-43). Palgrave Macmillan.

Statista Market Insights. (2023). *Gaming Revenue Forecasts by Market.* Statista.

Bai, D., Chen, L., Shang, Z., & Wang, Y. (2022). *Esports Industry, Video Game Industry and Economic Growth: An Empirical Research in China.* SSRN.

I.V., L., & M.V., P. (2021). The video game industry as an object and method for the development of modern culture. *World of Science. Series: Sociology, Philology, Cultural Studies*, No 4, Vol 12.

McKenzie, T., Trujillo, M. M., & Hoermann, S. (2019). Software Engineering Practices and Methods in the Game Development Industry. *CHI PLAY '19 Extended Abstracts: Extended Abstracts of the Annual Symposium on Computer-Human Interaction in*

Play Companion Extended Abstracts (pp. 181-193). Barcelona, Spain: Association for Computing Machinery.

McKenzie, T., Morales-Trujillo, M., Lukosch, S., & Hoermann, S. (2021). Is Agile Not Agile Enough? A Study on How Agile is Applied and Misapplied in the Video Game Development Industry . *ACM/IEEE International Conference on Global Software Engineering* (pp. pp. 94-105). Madrid, Spain: IEEE.

Arafat, M., Qusef, A., & Al-Taher, S. (2019). Steam's Early Access Model: A Study on Consumers' Perspective. *IEEE Jordan International Joint Conference on Electrical Engineering and Information Technology* (pp. 336-341). Amman, Jordan: IEEE.

Consalvo, M., & Paul, C. A. (2018). 'If you are feeling bold, ask for $3': Value Crafting and Indie Game Developers. *Transactions of the Digital Games Research Association* , pp. 51-75.

Piller, F., Ihl, C., Fuller, J., & Stotko, C. (2004). Toolkits for open innovation - the case of mobile phone games . *Annual Hawaii International Conference on System Sciences (HICSS)* (p. 10). Big Island, HI, USA: IEEE.

Erickson, K. (2018). Can creative firms thrive without copyright? Value generation and capture from private-collective innovation . *Business Horizons*, 699-709.

Burger-Helmchen, T., & Cohendet, P. (2011). User Communities and Social Software in the Video Game Industry. *Long Range Planning*, 317-343.

Wiederhold, B. K. (2021). Kids Will Find a Way: The Benefits of Social Video Games. *Cyberpsychology, Behavior, and Social Networking*, 213-214.

Wirman, H. (2014). Gender and Identity in Game-Modifying Communities . *Simulation & Gaming*, 70 - 92.

Gandolfi, E., Ferdig, R. E., & Soyturk, I. (2023). Exploring the learning potential of online gaming communities: An application of the Game Communities of Inquiry Scale . *New Media & Society*, 1374 - 1393 .

Salovaara, A., Johnson, M., Toiskallio, K., & Tiitta, S. (2005). Playmakers in multiplayer game communities: their importance and motivations for participation. *ACM SIGCHI International Conference on Advances in computer entertainment technology* (pp. 334–337). Valencia, Spain: Association for Computing Machinery.

Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013, January). Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology. *Organizational Research Methods*, pp. 15 - 31.

Adams, W. C. (2015). Conducting Semi-Structured Interviews. In Wiley, *Handbook of Practical Program Evaluation, Fourth* (pp. 492-505). Kathryn E. Newcomer; Harry P. Hatry; Joseph S. Wholey: Kathryn E. Newcomer, Harry P. Hatry, Joseph S. Wholey.

Magnani, G., & Gioia, D. (2023, April). Using the Gioia Methodology in international business and entrepreneurship research. *International Business Review*, pp. Volume 32, Issue 2.

Walt, J. V., Kroon, J., & Fourie, B. (2004, June). The importance of a vision and mission for small, medium-sized and large businesses. *South African Journal of Economic and Management Sciences*, pp. Volume 7, Issue 2.

Smale, S. D., Kors, M. J., & Sandovar, A. M. (2019, June). The Case of This War of Mine: A Production Studies Perspective on Moral Game Design . *Games and Culture*, pp. 387 - 409.

Arslan, I. K. (2020). THE IMPORTANCE OF CREATING CUSTOMER LOYALTY IN ACHIEVING SUSTAINABLE COMPETITIVE ADVANTAGE. *Eurasian Journal of Business and Management*, pp. 11-20.

Buell, R. W. (2019, April). Operational Transparency. *Harvard Business Review*, pp. 102-113.

Stevens, J. L., B. I., Breazeale, M., & Jones, C. L. (2018, May). Timeliness, transparency, and trust: A framework for managing online customer complaints. *Business Horizons*, pp. 375-384.

Ruggles, C., Wadley, G., & Gibbs, M. R. (2015). Online Community Building Techniques Used by Video Game Developers. *International Conference on Entertainment Computing* (pp. 114-125). Sanda, Japan: ICED.

Beedle, M., Bennekum, A. V., & Cockburn, A. (2001). *Manifesto for Agile Software Development.* Retrieved from The Agile Manifesto: http://www.agilemanifesto.org/

Alsaqqa, S., Sawalha, S., & Abdel-Nabi, H. (2020, July). Agile Software Development: Methodologies and Trends. *International Journal of Interactive Mobile Technologies*, pp. 246-270.

Najihi, S., & Elhadi, S. (2022). Software Testing from an Agile and Traditional view. *International Conference on Future Networks and Communications* (pp. 775-782). Ontario, Canada: Procedia Computer Science.

Larman, C. (2004). *Agile and Iterative Development: A Manager's Guide.* Addison-Wesley Professional.

Denning, S. (2013, March). Why Agile can be a game changer for managing continuous innovation in many industries. *Strategy & Leadership*, pp. 5-11.

Abrahamsson, P., Salo, O., & Ronkainen, J. (2002). *Agile Software Development Methods: Review and Analysis.* Espoo, Finland: VTT.

Rubio, N., Villaseñor, N., & Yagüe, M. J. (2021). Value co-creation between consumers and distributors: the moderating effect of relationship characteristics. *International Journal of Retail & Distribution Management*, 1067-1088.

Wadeisa, D., & Sada, A. (2015). *Innovation Through Co-creation: Strategies To Manage The Challenges Of Co-Creation.* Umeå, Sweden: Umeå School of Business and Economics.

Iglesias, O., Ind, N., & Alfaro, M. (2013, July). The organic view of the brand: A brand value co-creation model. *Journal of Brand Management*, pp. 670-688.

Cornelio, G. S., & Cruz, E. G. (2014). Co-creation and Participation as a Means of Innovation in New Media: An Analysis of Creativity in the Photographic Field. *International Journal of Communication*, pp. 1 - 20.

Prockl, G., Bhakoo, V., & Wong, C. (2017, March). Supply chains and electronic markets - impulses for value co-creation across the disciplines. *Electronic Markets*, pp. 135-140.

Priante, A., & Ehrenhard, M. (2017, December). Identity and collective action via computer-mediated communication: A review and agenda for future research. *New Media & Society*, pp. 2647 - 2669.

Piller , F., Ihl, C., & Füller, J. (2004). games, Toolkits for open innovation - the case of mobile phone. *Hawaii International Conference on System Sciences* (p. 10). Hawaii: IEEE.

Erickson, K. (2018, September). Can creative firms thrive without copyright? Value generation and capture from private-collective innovation . *Business Horizons*, pp. 699-709.

Erickson, K. (2018, September). Can creative firms thrive without copyright? Value generation and capture from private-collective innovation . *Business Horizons*, pp. 699-709.

Constantinides, E., Brünink, L., & Romero, C. L. (2015, April). Customer motives and benefits for participating in online co–creation activities. *International Journal of Internet Marketing and Advertising*, pp. 21-48.

Priharsari, D., Abedin, B., & Mastio, E. (2020, January). Value co-creation in firm sponsored online communities: What enables, constrains, and shapes value. *Internet Research*, pp. 763-788.

Nohutlu, Z. D., & Englis, E. (2023, November). Innovating With the Customer: Co-Creation Motives in Online Communities. *International Journal of Electronic Commerce* , pp. 523-557.

Urban, G. L. (2005, April). Customer Advocacy: A New Era in Marketing? *Journal of Public Policy & Marketing*, pp. 155 - 159.

Hajli, N., Shanmugam, M., Papagiannidis, S., Zahay, D., & Richard, M.-O. (2017, January). Branding co-creation with members of online brand communities. *Journal of Business Research*, pp. 136-144.

Zwass, V. (2010, December). Co-Creation: Toward a Taxonomy and an Integrated Research Perspective. *International Journal of Electronic Commerce*, pp. 11-48.

Prahalad, C., & Ramaswamy, V. (2004, June). Co-creating unique value with customers. *Strategy & Leadership*, pp. 4-9.

Krishnamurthy, S. (2005). An Analysis of Open Source Business Models. In J. Feller, *Perspectives on Free and Open Source Software.* The MIT Press.

Abelein, U., Sharp, H., & Paech, B. (2013). Does Involving Users in Software Development Really Influence System Success? *IEEE Software*, 17 - 23.

von Krogh, G., Haefliger, S., Spaeth, S., & W. Wallin, M. (2012). Carrots and Rainbows: Motivation and Social Practice in Open Source Software Development. *MIS Quarterly*, 649-676.

Bettenburg, N., & E. Hassan, A. (2013). Studying the impact of social interactions on software quality. *Empirical Software Engineering*, 375–431.

Vaisnore , A., & Petraite, M. (2011). Customer Involvement into Open Innovation Processes: a Conceptual Model. *Social sciences*, 62-73.

Chesbrough, H., & Crowther, A. K. (2006). Beyond high tech: early adopters of open innovation in other industries. *R&D Management*, 229-236.

Ye, Y., Yamamoto, Y., & Kishida, K. (2004). Dynamic community: a new conceptual framework for supporting knowledge collaboration in software development. *Asia-Pacific Conference on Software Engineering* (pp. 472-481). Busan, Korea: IEEE.

Howe, J. (2006). *The Rise of Crowdsourcing.* Retrieved from Wired.: https://www.wired.com/2006/06/crowds/

Lakhani, K., Garvin, D., & Lonstein, E. (2012, February). TopCoder (A): Developing Software through Crowdsourcing. *Harvard Business School General Management Unit* .

Mao, K., Capra, L., & Harman, M. (2017, April). A survey of the use of crowdsourcing in software engineerin. *Journal of Systems and Software*, pp. 57-84.

Schneider, C., & Cheung, T. (2013). The Power of the Crowd: Performing Usability Testing Using an On-Demand Workforce. *Information Systems Development.* New York, NY: Springer.

Mollick, E. (2014). The dynamics of crowdfunding: An exploratory study. *Journal of Business Venturing*, 1-16.

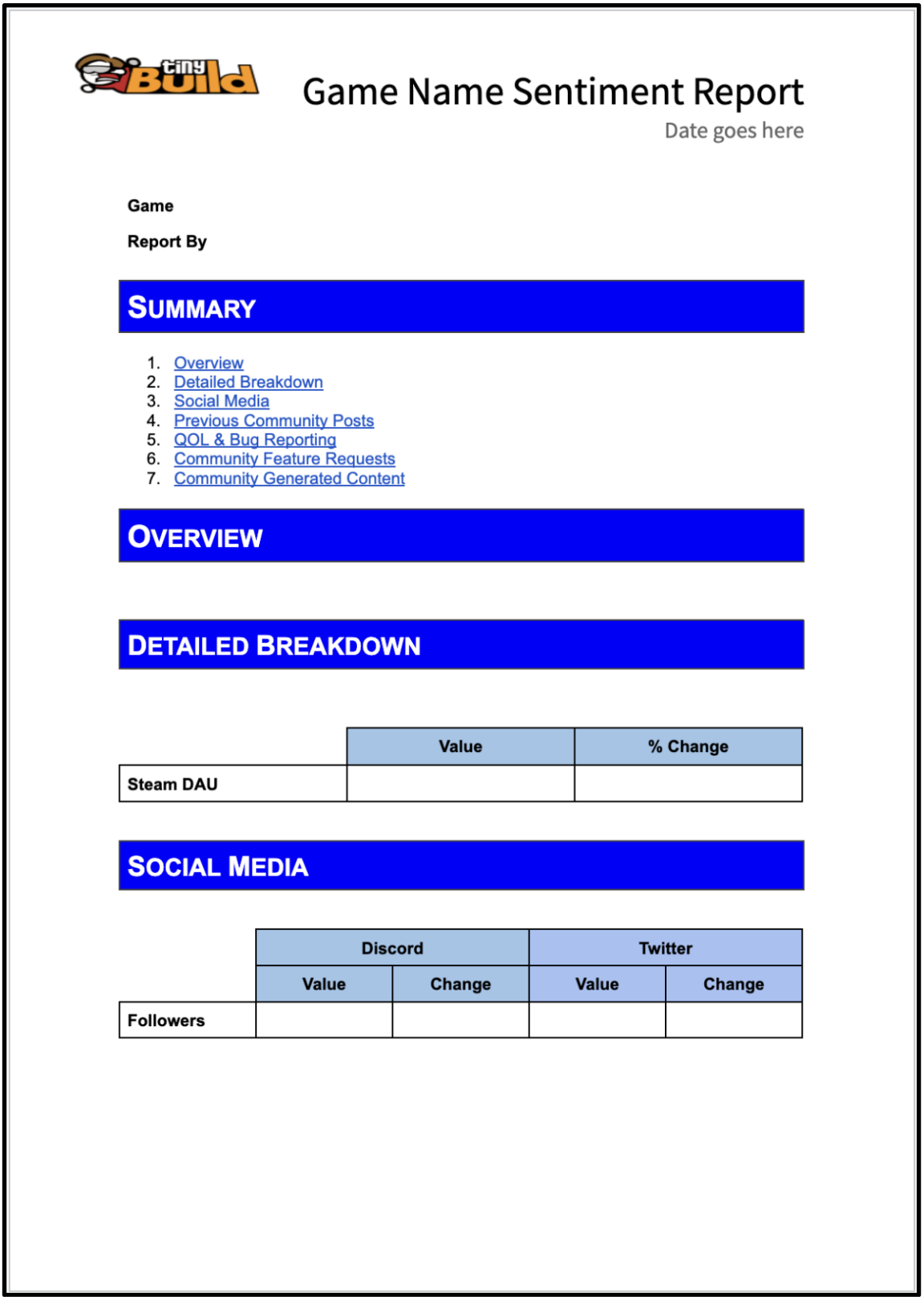
Branzov, T., & Maneva, N. (2014). Branzov, Todor, and Nelly Maneva. "Crowdfunding business models and their use in software product development. *International Scientific Conference Informatics in Scientific Knowledge*, 781-790.

Boylan, D. H., Nesson, D., & Philipps, J. (2018). Understanding crowdfunding for business funding – a legal and platform review. *Journal of Accounting & Organizational Change*, 273-290.

Schlãœter, A., Waldkirch, M., & Auernhammer, J. (2021, November). No Second Chance For A First Impression: The Role Of Aesthetics In Early Access Video Games. *International Journal of Innovation Management*, pp. Vol. 25, No. 10.

FIGURE 2.3-1: CORE PRINCIPLES OF THE AGILE FRAMEWORK (BEEDLE, BENNEKUM, & COCKBURN, 2001) 4
FIGURE 2.4-1: EXAMPLES OF CUSTOMER BEHAVIOURS AND COMPANY PRACTICES (NOHUTLU & ENGLIS, 2023) 7
FIGURE 2.4-2: COMBINATIONS OF THE DART MODEL (PRAHALAD & RAMASWAMY, 2004) 8
FIGURE 2.4-3: RECURSIVE PROCESS OF BRANDING CO-CREATION (HAJLI, SHANMUGAM, PAPAGIANNIDIS, ZAHAY, & RICHARD, 2017) 9
FIGURE 2.5-1: INTERSECTION BETWEEN CROWDSOURCING AND SOFTWARE DEVELOPMENT (MAO, CAPRA, & HARMAN, 2017) 11
FIGURE 2.5-2: VARIANTS OF CROWDFUNDING (BRANZOV & MANEVA, 2014) 12
FIGURE 2.6-1: CORE PILLARS BASED ON LITERATURE 14
FIGURE 3.1-1: SUMMARY OF CANDIDATE DEMOGRAPHICS 16
FIGURE 3.2-1: SAMPLE OF INITIAL CONCEPTS FROM INTERVIEWS 16
FIGURE 3.2-2: CONCEPTS GROUPED INTO EMERGING THEMES 16
FIGURE 3.2-3: OVERVIEW OF GIOIA METHOD 17
FIGURE 3.2-4: EXAMPLE OF SUCCESS FACTOR ANALYSIS 17
FIGURE 4.2-1: VISUALISING TRADE-OFF BETWEEN CONTENT & REFINEMENT 20
FIGURE 4.2-2: COMMON PHASES OF SELECTIVE AND PUBLIC ACCESS 22
FIGURE 4.3-1:INTERNAL AND EXTERNAL IMPACT OF PRODUCT VISION 23
FIGURE 4.6-1: CONVENTIONAL FEEDBACK LOOP 30
FIGURE 4.6-2: EVOLVED FEEDBACK LOOP FOR EARLY ACCESS 31
FIGURE 4.6-3: METHODS FOR INFORMATION DISPERSION 32
FIGURE 4.6-4: METHODS FOR COLLECTING FEEDBACK AND INSIGHTS 33
FIGURE 4.8-1: PROPOSED UPDATED FRAMEWORK 35

8.1. User Sentiments Report



Game Name Sentiment Report

Date goes here

Game

Report By

SUMMARY

1. Overview
2. Detailed Breakdown
3. Social Media
4. Previous Community Posts
5. QOL & Bug Reporting
6. Community Feature Requests
7. Community Generated Content

OVERVIEW

DETAILED BREAKDOWN

	Value	% Change
Steam DAU		

SOCIAL MEDIA

	Discord		Twitter	
	Value	Change	Value	Change
Followers				

Game Name Sentiment Report

Date goes here

Game

Report By

SUMMARY

1. Overview
2. Detailed Breakdown
3. Social Media
4. Previous Community Posts
5. QOL & Bug Reporting
6. Community Feature Requests
7. Community Generated Content

OVERVIEW

DETAILED BREAKDOWN

	Value	% Change
Steam DAU		

SOCIAL MEDIA

	Discord		Twitter	
	Value	Change	Value	Change
Followers				

Game Name Sentiment Report

Date goes here

COMMUNITY GENERATED CONTENT

Content Summary	Link

8.2. Interview Questions

Full Interview Questions:

Introduction, definition and Background:

- Can you please introduce yourself, your background, and your role within the organisation?
- Can you please give your definition of Early Access, and what different methodologies, practices and operations fall under this term?
    - In your opinion, what thoughts would you have on the share of Early Access games in the industry? Has this share changed in recent years?
- When referring to the label of "Early Access", has the meaning of the term evolved over the years?
    - If so, do you have any hypothesis on how it will evolve in the future?

Exploring Experience with Early Access:

- Have you had any experience working with titles that leveraged early access?
    - If yes, could you share some details about how you leveraged the EA model?
- What type of activities or initiatives were included in this process?
    - Can you share some examples or insights from these activities and their results?
    - What impact did these activities have on the company, game release and community?
- From your experience, are there any specific scenarios that are more tailored to early access models?
    - In your experience, does it enable any new segments or opportunities?
    - Does it imply new challenges and threats that previously weren't a consideration?

Process Considerations for Early Access:

- How does the EA model impact the overall design and development process?
    - Can you give us an overview of the process and how it differs from other development models used in game development?
    - Are there any specific scenarios that are more tailored to early access models?
    - In your experience, does it enable any new options or opportunities?
- Are there any benefits or drawbacks to using this model over others?
    - Have you encountered any challenges or negative experiences with early access models? How do you address them?

User Feedback and Iterative Development:

- What role do the users play in the design and development activities? How influential can user feedback be in the final product?
    - How is user involvement factored into the process? And how much impact can users have on the final product?
    - What aspects of the design and development process can users affect, and what influence can they have in those areas?
    - Have there been any instances where user feedback significantly influenced the direction or features of the product?
    - How do you handle feedback or suggestions that may not align with the development vision or roadmap?
- How do you go about measuring and tracking user involvement?
    - Are there any key metrics or processes used to track user involvement in the process?

    - Are there any specific methods and tools that are used to translate user feedback into design and development decisions?
    - How do you estimate the number of users needed for a relevant amount of feedback?
- Are there any steps or activities that can help with increasing user engagement for specific aspects, such as design generation?
    - Do you have any specific initiatives or activities to encourage user interactions and collaboration within the early access community?

Managing Expectations and Competition:

- Can you share some details of your experience with managing user expectations during the EA phase?
    - How do you manage user expectations during the early access phases, considering that the product may still be under development?
    - How influential is the pricing structure in the expectations from users? How did you determine the structure to follow?
- How do you evaluate and mitigate the risks associated with releasing a partially developed product to the public?
    - Are there any specific contingency plans or strategies in place to address unexpected challenges or issues that may arise from early access?
- How do you assess the competitive landscape and positioning of a product when considering early access?
- Have you observed any specific market niches or genres where early access can be particularly advantageous for differentiation?

Learning from Early Access

- What lessons have you learned from previous experiences with early access models?
    - Have there been any unexpected benefits or insights gained through the early access process?
- How do you apply the knowledge and feedback obtained from early access to future product development and decision-making?

Studio Reputation and it's Impact:

- Does the reputation of a studio influence the early access process in any way? What kind of impact can this have?
    - What steps do you take to minimise potential negative impacts on the brand or reputation during the early access phase?

Closing

- Is there anything else you would like to share or discuss early access models and user involvement in development activities?
- Thank the interviewee for their time and participation in the interview.

8.3. Candidate Summary

#	Role	Focus Area	Experince/ Backgound	Job Function	Location	Number of Releases	Company Size (Employees)	Genre Specialization
1	community manager	communication	communty, production	community management, feedback magement	Sweden	4	40	MOBA, Survival
2	community manager	communication	community moderator	community management	Norway	4	<25	platform, rouguelike
3	community manager	development	developer	level designer, cm	Sweden	5	40	strategy, multiplayer, survival, fps
4	community developer	communication, management	developer	community management, feedback magement	France	6	200	strategy
5	Senior community manager	communication, development, engagement	communty, production	community management	Canada	2	30	MOBA, Exploration
6	developer/ community developer	communication, tecnichal implementations	communty, production	community, cloud infrastructure,	New Zealand	4	50	strategic, survival, fps
7	associate producer	communication, animations	communty, production	community management	France	5	50	roguelike
8	community manager / publishing	communication, publishing	streaming, events, q&a	community management, feedback magement	USA	3 (60 as publisher)	150	various
9	cluster manager / managing director	management, development	development, business development	cluster management	Sweden	20 as a publisher	<50	various
10	creative director	management, development	High level game design, business development	founder	USA	3	30	roguelike
11	developer/ community developer	management, development	communty, production, development	various	Czech republic	29	400	strategy, simulation, fps
12	community manager	communication	community moderator	community management	USA	100 as publisher	200	various

TRITA – ITM-EX 2024:167
Stockholm, Sweden 2024

www.kth.se